

RECOVERY OF INFORMATION METRICS: MULTIPLICATION FAILURE SCHEMES

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ABSTRACT. The second jet of a squared-distance function determines a normal compression of the inverse information matrix. For a calibrated Hankel family, its ambiguity is dual to failure of quadratic multiplication of normal polynomials. We determine the codimension of this failure locus for independently designed subspaces. In a stable dimension range we classify the entire determinantal scheme when it has expected codimension: it is integral and Cohen–Macaulay in one regime, and reduced with one nondegenerate component and 2^{L-1} signed secant components at the wall. In the excess regime the signed secant varieties are precisely the maximal-dimensional components; they are rational and generically smooth in the determinantal scheme. Removing the two secant points identifies the wall with saturation of a residual multiplication map. We obtain the fundamental cycle in expected codimension and local small-singular-value exponents at smooth real native strata. An exact positive realization criterion transfers these results to synchronized quadratic images and yields the sharp single-contact recovery threshold. The physical boundary experiment, quantitative Poisson comparison, and a deterministic finite-precision score reduction connect multiplication rank to local information loss. Finite-sample estimation and complementary contact results are retained in the appendices.

1. INTRODUCTION

Let $M \subset \mathbb{R}^k$ be a known residual image and $R > 0$ an inverse information matrix. Consider

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b). \quad (1.1)$$

At a smooth point, its full second jet sees only the normal compression of R . A statistical model may constrain R enough to remove this ambiguity. The inverse problem is therefore governed both by the geometry of M and by the linear space of admissible information matrices.

We study the calibrated family

$$R_\theta = JH(\theta)J^\top, \quad H(\theta)_{ab} = \theta_{a+b}, \quad 0 \leq a, b \leq k-1,$$

where J is fixed and invertible, and θ lies in an open positive moment cone of dimension $p = 2k-1$. Set $n = k-1$ and $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$. Dehomogenization identifies real coefficient vectors with $\mathcal{P}_n$. A normal space N corresponds to $U = J^\top N$. The observations specify the moment functional on $U^2 = \text{span}\{fg : f, g \in U\}$. Multiple contacts specify a sum of such spaces, without cross-contact products.

The central geometric issue is not only whether a general product map is onto, but how large its entire failure locus is. Put $q = \binom{c+1}{2}$ and

$$X_L = \text{Gr}(c, V_n)^L, \quad \Delta^{(L)} = \{(U_1, \dots, U_L) : \sum_\nu U_\nu^2 \neq V_{2n}\}.$$

All codimensions in the following theorem are complex codimensions of reduced closed sets.

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Theorem 1.1 (Reduced failure locus and independent contacts). *Suppose $4 \leq c < k$, $L \geq 1$, and $Lq \geq p$. Then*

$$\text{codim}_{X_L} \Delta^{(L)} = \min\{Lq - p + 1, Lc - 3\}. \quad (1.2)$$

There is an irreducible common-secant subvariety $\Sigma_L \subset \Delta^{(L)}$ of codimension $Lc - 3$. If $Lc - 3 \leq Lq - p + 1$, it is a maximal-dimensional irreducible component of $\Delta^{(L)}$. Consequently a general L -tuple is informative if and only if $Lq \geq p$.

The common-secant condition means that all U_ν lie in a hyperplane represented by a point of the secant variety of the evaluation rational normal curve in $\mathbb{P}(V_n^*)$. Thus the second term of (1.2) is a concrete geometric obstruction, not a formal expected codimension. For one contact, the common-base-point locus lies properly inside this larger irreducible family and is not itself a component of the failure locus.

The numerical formula has a structural refinement. Write $a = Lq - p + 1$ and $b = Lc - 3$. In the stable range $c \geq 8$, $k \geq 2c + 1$, Theorem 5.1 proves that the maximal-minor scheme is integral and Cohen–Macaulay when $a < b$. Its generic annihilator is a nondegenerate Hankel form. At $a = b$ the scheme is reduced and Cohen–Macaulay, with precisely one nondegenerate component and 2^{L-1} signed secant components. If $b < a$, the latter are exactly the maximal-dimensional components. They are rational and generically smooth in the determinantal scheme. The signs record independent choices of the two isotropic hyperplanes of a rank-two form, modulo their simultaneous interchange; the common-secant family above is only the all-equal choice.

The change of regime has an intrinsic residual description. At two distinct secant points, write $U_\nu \cap \text{rad } H_\ell = gK_\nu$, where g is their vanishing quadratic. The wall is

$$L \binom{c}{2} = 2k - 5,$$

exactly the saturation threshold for multiplication of the $(c - 1)$ -planes K_ν in the $(k - 2)$ -dimensional residual space. This identity determines the conormal dimension of a general signed component. The expected-codimension cycle and local conditioning exponents follow from the structural theorem. The earlier all-dimension codimension formula is not restricted by the additional hypotheses of this refinement.

For the synchronized residual image

$$q_A(\eta) = ((a_i^\top \eta)^2)_{i=1}^k, \quad A \in \mathbb{R}^{k \times d},$$

the normal dimension is $c = k - d$. Native realizability is an open positive-vector condition. Global separation at a prescribed normal space requires, in addition, that the corresponding tangent plane not split across a nontrivial coordinate partition. This exact criterion is proved in Theorem 6.1.

Theorem 1.2 (Sharp native single-contact threshold). *Fix $d \geq 2$ and any separated root and clock configuration of Section 3. For $k > d$, there exist full-rank loadings A with $q_A(e_1) > 0$ for which the single full second jet $D^2 C_{R_\theta}(q_A(e_1))$ identifies every $\theta \in \Omega$ if and only if*

$$\binom{k - d + 1}{2} \geq 2k - 1.$$

In the sufficient range the assertion holds on a nonempty Zariski-open loading set, for the global distance function. In particular

$$k_{\min}(d) = d + \left\lceil \frac{3 + \sqrt{16d + 1}}{2} \right\rceil = d + 2\sqrt{d} + O(1). \quad (1.3)$$

Unrestricted positive matrices at the same contact retain an invisible space of dimension $d(2k - d + 1)/2$.

A contact here is a matrix-valued observation with q entries, not a scalar datum. The comparison in (1.3) is across calibrated families with varying k , fixed d , and arbitrary separated configurations at each k ; it is not a universal threshold for every information model. Independently designed contacts can use different loadings with the same J . Several points of one fixed loading are not asserted to vary independently.

Method and relation to earlier theory. Hankel duality with polynomial multiplication is classical [13]. The saturated one-space maximal-rank statement is also a consequence of classical postulation of general rational curves [2, 3, 12]. Our proof of Theorem 1.1 instead combines the classical dimensions of Hankel rank strata [5, Sections 1–2] with an isotropic-incidence count, an optimization over all ranks, and an explicit common-secant family. This proves the full codimension formula and the coupled multi-space assertion, not just generic surjectivity. The classical maximal-minor height bound supplies the other matching upper bound [6, Introduction]. These classical inputs are distinguished from the incidence optimization and native fixed-space realization proved here.

Failure of quadratic normality has a substantial antecedent literature. Green–Lazarsfeld [19, Theorems 1 and 3] study complete very ample linear series on curves, giving normal-generation criteria and finite secant configurations explaining failure under degree and cohomology hypotheses. Ballico [18, Theorem 0.2] studies failure cycles for a fixed embedded surface and detects failure on finite linear sections, using a Grassmannian incidence and cohomology. The principle that secants can explain quadratic-normality failure is therefore classical. Here the complete series on $\mathbb{P}^1$ is fixed and normally generated; the varying objects are its proper c -dimensional subseries, and for $L > 1$ the observation uses a direct sum of within-subseries products, without cross-products. Our structural assertion concerns the components and generic scheme structure over this product of Grassmannians. Neither the complete-series degree criterion nor the surface-section failure-cycle theorem supplies this signed component classification or the residual wall. We distinguish these particular conclusions from the classical inputs on Hankel ranks, determinantal resolutions, and cycles [5, 20, 21, 22, 23].

Polynomial product spaces also occur in the study of Gram spectrahedra [15]; the present observation is the dual moment functional on within-contact products. The monomial specialization is a restricted additive-basis problem [17, 10, 11], and can have a strictly worse threshold. Quadratic sensing and phase retrieval concern reconstruction of the input to a quadratic measurement map [1]; our unknown is instead an information metric, and injectivity on all of Sym_d is not assumed. Complete-conormal projective duality [8] addresses a different, larger observation category.

The statistical statements likewise keep their experiments separate. The coalesced-root model has a smooth positive extension, but its physical local parameter set is a cone. Its efficient Gaussian quotient and $N^{-1/4}$ root-amplitude scale are derived explicitly in Section 9. For observed Poisson exposures we form normal-compression score statistics directly from the counts. Their local information loss is exactly the loss of rank of the product map. This is a local likelihood statement in a specified target subexperiment, not a claim of global equivalence of all raw data and finitely many distance readings. The finite-sample estimators and computed finite-offset features in the appendices are retained as distinct procedures. Standard constrained and regular asymptotic principles are attributed to [7, 16].

2. EXACT FIBRES OF PRESCRIBED CONTACT DATA

For $\theta \in \mathbb{R}^p$ write

$$\ell_\theta \left(\sum_{a=0}^{2n} f_a t^a \right) = \sum_a f_a \theta_a.$$

Let $\Omega \subset \mathbb{R}^p$ be open with $R_\theta > 0$. At each prescribed point x_ν , assume that the global image has one embedded smooth germ and that its local minimizing projection computes (1.1). Parameter differentiation below uses analytic germs and a common projection tube on compact metric sets. These conditions are verified for the native loading family in Section 6.

Let Z_ν have orthonormal columns spanning the Euclidean normal space. Put

$$B_\nu = Z_\nu^\top R_\theta Z_\nu, \quad G_\nu = \frac{1}{2} Z_\nu^\top D^2 C_{R_\theta}(x_\nu) Z_\nu.$$

Lemma 2.1 (Normal block). *One has $G_\nu = B_\nu^{-1}$. The full Hessian has tangent kernel and is determined by this normal block.*

Proof. The quadratic contact with a manifold equals that with its tangent space T . At the minimizer of $(u - v)^\top R^{-1}(u - v)$ over $v \in T$, the vector $R^{-1}(u - v)$ is normal. Hence $u - v = RZn$, and normal projection gives $Bn = Z^\top u$. The minimum is $(Z^\top u)^\top B^{-1} Z^\top u$. A local parametrization and its positive tangent Gram matrix justify the minimizing projection by the implicit function theorem and yield this quadratic term. $\square$

Set $U_\nu = J^\top \text{ran } Z_\nu$ in polynomial coordinates, and define

$$\mathcal{U} = \text{im} \left(\bigoplus_{\nu=1}^L \text{Sym}^2 U_\nu \xrightarrow{\sum \text{multiplication}} \mathcal{P}_{2n} \right), \quad \mathcal{L}\theta = (B_\nu(\theta))_\nu.$$

The codomain of $\mathcal{L}$ has the direct-sum Frobenius norm.

Theorem 2.2 (Exact contact fibre). *The exact second-jet fibre through θ is*

$$(\theta + \mathcal{U}^\perp) \cap \Omega.$$

Its local dimension is $p - \dim \mathcal{U}$. Recovery on Ω is equivalent to $\mathcal{U} = \mathcal{P}_{2n}$, or to injectivity of $\mathcal{L}$.

Proof. For normal vectors v, w with coefficient polynomials f_v, f_w ,

$$v^\top (R_{\theta'} - R_\theta) w = \ell_{\theta' - \theta}(f_v f_w).$$

Equality of the jets is equality of the positive normal blocks, by Lemma 2.1. It is therefore precisely annihilation of $\mathcal{U}$. Inversion is a diffeomorphism on positive blocks; openness of Ω gives the asserted fibre dimension. $\square$

For one d -dimensional germ there are at most $q = \binom{k-d+1}{2}$ independent products. For unrestricted matrices, fixing the normal-normal block leaves tangent-tangent and mixed blocks free, with total dimension

$$\frac{k(k+1) - (k-d)(k-d+1)}{2} = \frac{d(2k-d+1)}{2}. \quad (2.1)$$

Small perturbations in every such direction preserve positivity.

3. THE NATIVE INFORMATION FAMILY

Fix distinct strictly positive rank-one four-cell probability tables M_1, M_2 , roots $0 < r_1 < \dots < r_k < D$, and $m = 2k + 1$ clocks $D < T_1 < \dots < T_m$. Use

$$f_1(T) = f_{1,0}(T) \prod_i ((T - r_i - s_i/2)^2 - v_i), \quad p_j = \pi_j M_1 + (1 - \pi_j) M_2,$$

$$\frac{\pi_j}{1 - \pi_j} = \frac{\alpha f_1(T_j)}{(1 - \alpha) f_2(T_j)}.$$

The fixed factors are positive at the clocks. At $s = v = 0$, the free coordinates are log mixing odds and the root-sum displacements; the physical retained coordinates satisfy $v_i \geq 0$. The likelihood is positive and analytic on a two-sided neighbourhood of this point. Its restriction to $v \geq 0$, not that extension's unrestricted estimator, is the physical experiment.

Let $\lambda_j > 0$ be information weights and set

$$\kappa_j = \pi_j^2 (1 - \pi_j)^2 \sum_{a=1}^4 \frac{(M_{1a} - M_{2a})^2}{p_{ja}}, \quad V_j = (1, -((T_j - r_i)^{-1})_i, -((T_j - r_i)^{-2})_i).$$

The k simple poles precede the k double poles. Write

$$h(t) = \prod_i (t - r_i)^2, \quad p_*(t) = \prod_i (t - r_i), \quad P(t) = \prod_j (t - T_j),$$

$$d_i = \frac{P(r_i)}{\prod_{a \neq i} (r_i - r_a)^2}, \quad \xi_j = \frac{h(T_j)}{P'(T_j)},$$

$$J_{ia} = [t^a] d_i \frac{p_*(t)}{t - r_i}, \quad \rho_j = \frac{\xi_j^2}{p_*(T_j)^2}. \quad (3.1)$$

Proposition 3.1 (Profiled quadratic information). *The retained block of the inverse full information, equivalently the inverse nuisance Schur complement, is*

$$R = JH(\theta)J^\top, \quad \theta = \sum_{j=1}^m \frac{\rho_j}{\lambda_j \kappa_j} w_{2n}(T_j), \quad w_s(t) = (1, t, \dots, t^s)^\top. \quad (3.2)$$

The matrix J is invertible. Independent positive information weights give an open moment cone Ω of dimension $p = 2k - 1$.

Proof. Multiplying a linear combination of the columns of V by h gives a polynomial of degree at most $2k$. Vanishing at the $2k + 1$ clocks forces it to be zero; principal parts at the roots and the constant at infinity then force every coefficient to vanish. Thus V is invertible and the full information is $V^\top \text{diag}(\lambda_j \kappa_j) V$.

For arbitrary log-odds increments y_j , interpolate

$$g(t) = \sum_j h(T_j) y_j \frac{P(t)}{(t - T_j) P'(T_j)}.$$

The retained coefficient at r_i is $-g(r_i) / \prod_{a \neq i} (r_i - r_a)^2$. Thus the retained rows Z_0 of V^{-1} are

$$(Z_0)_{ij} = \frac{\xi_j d_i}{T_j - r_i} = \frac{\xi_j}{p_*(T_j)} (J w_n(T_j))_i.$$

Their weighted Gram matrix gives (3.2). Evaluation of the polynomials defining J at the distinct roots is diagonal with nonzero entries, proving invertibility. The positive moment weights range over an open orthant. The moment matrix has rank p by Vandermonde independence, so its image is open. Positive atomic weights at distinct clocks give positive definite $H(\theta)$. $\square$

The weights in this proposition need not be unknown likelihood parameters. For a realized fixed allocation they are n_j/N ; for a limiting allocation they are its limits; a different prespecified λ defines a target design metric. Section 10 uses a further protocol in which exposures are unknown but their counts are observed. Section 9 identifies (3.2) as the covariance of the efficient Gaussian quotient of the physical boundary experiment.

4. THE GLOBAL MULTIPLICATION FAILURE LOCUS

Throughout this section the field is $\mathbb{C}$, $k = n + 1$, $4 \leq c < k$, $q = \binom{c+1}{2}$, and $p = 2k - 1$. On $X_L = \text{Gr}(c, V_n)^L$ let $\mathcal{S}_\nu$ denote the tautological bundles. The failure scheme is defined by the maximal minors of

$$\mu : \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{S}_\nu \longrightarrow V_{2n} \otimes \mathcal{O}_{X_L}.$$

Its underlying set is $\Delta^{(L)}$. We shall prove Theorem 1.1 without using a maximal-rank theorem for general curves.

4.1. Annihilating Hankel forms. For $\ell \in V_{2n}^*$, let $H_\ell(f, g) = \ell(fg)$ on V_n . The following classical input concerns the whole rank locus, including confluent and limiting forms; it is not an assertion only about sums of distinct point evaluations.

Lemma 4.1 (Classical Hankel rank strata). *In $\mathbb{P}(V_{2n}^*)$, the rank-exactly- r stratum of H_ℓ is nonempty and has dimension*

$$h_r = \begin{cases} 2r - 1, & 1 \leq r < k, \\ 2k - 2, & r = k. \end{cases}$$

For $r < k$, its closure is the variety of sums of at most r point evaluations, with Zariski closure understood.

Identification with the classical statement. The rank- $\leq r$ equations are the $(r + 1)$ -minors of the $k \times k$ Hankel matrix in $2k - 1$ independent coordinates. The equality of Hankel minor ideals under changing the rectangular shape, recalled in [5, Section 1], identifies their coordinate ring with that of the maximal minors of an $(r + 1) \times (2k - r - 1)$ Hankel matrix. That ring has affine dimension $2r$. The secant identification is [5, Section 2, equation (2.0.1)]; thus the projective dimension is $2r - 1$. The rank- $\leq r - 1$ subvariety has smaller dimension, so the exact-rank stratum is nonempty and dense. The full-rank stratum is open in $\mathbb{P}^{2k-2}$. These are classical Hankel determinantal facts; no claim about their rational singularities is needed here. $\square$

Lemma 4.2 (Isotropic incidence dimensions). *Let a symmetric form on a k -space have rank r and radical E of dimension $k - r$. For a totally isotropic c -plane U , put*

$$s = \dim((U + E)/E), \quad a = c - s.$$

Such planes exist precisely for

$$\max\{0, c - k + r\} \leq s \leq \min\{c, \lfloor r/2 \rfloor\}. \quad (4.1)$$

The stratum with this value of s has codimension

$$D_{r,s} = (c - s)(r - s) + \frac{s(s + 1)}{2} \quad (4.2)$$

in $\text{Gr}(c, k)$.

Proof. Choose $U \cap E$, an a -plane in E , with dimension $a(k - r - a)$. The image of U in the nondegenerate r -dimensional quotient is an isotropic s -plane, whose Grassmannian has dimension $s(r - s) - s(s + 1)/2$. This follows by differentiating the symmetric restriction equations at an isotropic frame: pairing with a complementary dual frame gives all symmetric variations, so the equations are independent. Existence is equivalent to $2s \leq r$. Finally the lifts are an affine space of dimension $s(k - r - a)$. Subtracting the sum from $c(k - c)$ gives (4.2). Maximal orthogonal Grassmannians may have two components; both have this same dimension, which is all the count uses. $\square$

Lemma 4.3 (Rank optimization). *Assume $Lq \geq 2k - 1$. For all r, s satisfying (4.1),*

$$LD_{r,s} - h_r \geq \min\{Lq - 2k + 2, Lc - 3\}. \quad (4.3)$$

Proof. Write $A_0 = Lq - 2k + 2$ and $B_0 = Lc - 3$. If $r = k$, feasibility forces $s = c$, giving exactly A_0 . Suppose $r < k$. For $s = 0$, the left side is $(Lc - 2)r + 1 \geq Lc - 1 > B_0$.

For $1 \leq s \leq c - 2$, the coefficient of r in $LD_{r,s} - 2r + 1$ is $L(c - s) - 2 \geq 0$. Since $r \geq 2s$, the expression is at least

$$g(s) = 1 + s(Lc - 4) - \frac{Ls(s - 1)}{2}.$$

This is concave as a function of s , and

$$g(1) = B_0, \quad g(c - 2) - B_0 = (c - 3) \left(\frac{L(c + 2)}{2} - 4 \right).$$

The latter is nonnegative except when $(L, c) = (1, 4)$ or $(1, 5)$. In the first exception, saturation forces $k = 5$ and feasibility forces $s \leq 1$. In the second, $k \leq 8$; the values $g(1), g(2), g(3)$ are 2, 2, 1. The last value is feasible only when $k = 8$, in which case $A_0 = 1$. Both exceptions therefore satisfy (4.3).

For $s = c - 1$ and $L = 1$, the expression decreases with r , hence is at least its value at $r = k - 1$, namely $A_0 + k - 2c + 1$. Feasibility implies $k \geq 2c - 1$. For $s = c - 1$ and $L \geq 2$, it is minimized at $r = 2c - 2$ and exceeds B_0 by

$$(c - 2) \left(\frac{L(c + 1)}{2} - 4 \right) \geq 0.$$

Finally, if $s = c$ and $r < k$, it is $Lq - 2r + 1 \geq A_0 + 1$. These cases exhaust (4.1). $\square$

4.2. The common-secant family. Let $C_n \subset \mathbb{P}(V_n^*)$ be the evaluation rational normal curve, and let S_n be the closure of its secant lines. Since $n \geq 4$, S_n is irreducible of dimension three. Define

$$\mathfrak{S}_L = \{([h], U_1, \dots, U_L) : [h] \in S_n, U_\nu \subset \ker h\}, \quad \Sigma_L = \text{pr}_{X_L} \mathfrak{S}_L. \quad (4.4)$$

Proposition 4.4 (A global secant obstruction). *The closed set Σ_L is irreducible, has codimension $Lc - 3$ in X_L , and lies in $\Delta^{(L)}$.*

Proof. The incidence is a relative product of Grassmannians over S_n , so it is irreducible of dimension $3 + Lc(k - 1 - c)$ and its projection is closed. For a general $h = \alpha \text{ev}_x + \beta \text{ev}_y$, with $x \neq y$ and $\alpha\beta \neq 0$, every $f \in \ker h$ satisfies $\alpha f(x) = -\beta f(y)$. Consequently

$$\ell = \alpha^2 \text{ev}_x^{(2n)} - \beta^2 \text{ev}_y^{(2n)}$$

is nonzero and annihilates all products from each U_ν . Closedness extends this conclusion to the whole incidence.

It remains to check the dimension of the image, rather than merely that of the incidence. Fix a general h . If $c = k - 1$, the hyperplane U_1 determines h . Otherwise, the c -planes in $\ker h$ that are also annihilated by some $h' \in S_n \setminus \{h\}$ form a constructible set of dimension at most

$$3 + c(k - 2 - c) < c(k - 1 - c),$$

because $c \geq 4$. Here, for each h' , the plane lies in the intersection of two distinct hyperplanes. Thus a general $U_1 \subset \ker h$ admits no such h' , and the incidence projection is generically one-to-one. Subtraction from $\dim X_L = Lc(k - c)$ gives $Lc - 3$. $\square$

Proof of Theorem 1.1. Failure is equivalent to the existence of a nonzero ℓ annihilating every U_ν^2 . Consider the closed projective incidence of such pairs $((U_\nu), [\ell])$. On the rank- r stratum of H_ℓ , stratify each U_ν by s_ν . Lemmas 4.1 and 4.2 bound the incidence dimension by

$$\dim X_L + h_r - \sum_{\nu} D_{r, s_\nu}.$$

Its image cannot have larger dimension. For fixed r , each term is at least the smallest feasible $D_{r, s}$, so Lemma 4.3 gives

$$\text{codim } \Delta^{(L)} \geq \min\{Lq - p + 1, Lc - 3\} \geq 1.$$

This proves, in particular, generic surjectivity without assuming it.

Proposition 4.4 gives the upper bound $Lc - 3$ and shows that the failure set is nonempty. On a Grassmannian chart the map μ is a $p \times Lq$ matrix over a regular ring. The classical maximal-minor height bound gives codimension at most $Lq - p + 1$ for its nonempty failure locus; see [6, Introduction], with generic rank p and minor size p . The two upper bounds match the lower bound. When $Lc - 3$ is the minimum, the irreducible closed subvariety Σ_L has the maximal dimension of $\Delta^{(L)}$, so it is an irreducible component of its underlying reduced set. If $Lq < p$, the dimension of the domain of μ makes surjectivity impossible. This proves the final equivalence. $\square$

Corollary 4.5 (The base-point locus and the contact count). *For $L = 1$, the locus $\mathcal{B}$ of spaces with a common projective zero is irreducible of codimension $c - 1$, has multiplication corank at least two, and is properly contained in Σ_1 . It is not an irreducible component of $\Delta^{(1)}$. For arbitrary L , the least number of independently chosen c -planes with full product span is*

$$L_{\min} = \left\lceil \frac{2k - 1}{\binom{c+1}{2}} \right\rceil.$$

Proof. For a fixed linear factor l , common-base-point spaces form $\text{Gr}(c, lV_{n-1})$. Varying $[l] \in \mathbb{P}^1$ gives an irreducible incidence of dimension $1 + c(k - 1 - c)$, generically one-to-one because a general residual space has no further common factor. The codimension is $c - 1$. Products are divisible by l^2 , whose multiples have dimension $p - 2$. The annihilating hyperplane is an evaluation point of $C_n \subset S_n$, proving inclusion in Σ_1 . The codimensions $c - 1$ and $c - 3$ make that inclusion proper. An irreducible proper subset of the irreducible closed set $\Sigma_1 \subset \Delta^{(1)}$ cannot itself be a component of $\Delta^{(1)}$. The contact count follows from Theorem 1.1 and the dimension obstruction. $\square$

The theorem determines the exact dimension of the entire bad set and one explicit maximal-dimensional component in the stated regime. It does not assert that every component is a secant component, nor that the determinantal scheme is reduced, equidimensional, or everywhere transverse. In particular, the equality case of the two codimension bounds does not exclude additional components of the same dimension.

5. COMPONENTS, SCHEME STRUCTURE, AND THE RESIDUAL WALL

Let $\mathcal{D}_L$ denote the scheme of maximal minors of μ , and write

$$e = Lq, \quad a = e - p + 1, \quad b = Lc - 3, \quad m_X = Lc(k - c), \quad \mathcal{E} = \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{S}_\nu.$$

Thus $|\mathcal{D}_L| = \Delta^{(L)}$. The letters a, b in this section are codimensions, not loading vectors. Theorem 1.1 holds throughout its stated range. The structural results below assume the additional stable-range conditions

$$c \geq 8, \quad k \geq 2c + 1, \quad e \geq p. \quad (5.1)$$

For the classical input in Lemma 4.1, the precise affine dimension statement is $\dim R = 2t - 2$ on page 2 of [5]'s arXiv version; set $t = r + 1$. The secant identification is its equation (2.0.1). Exact-rank strata are open in their rank-at-most closures; confluent forms remain included. These inputs do not supply the component classification below.

These conditions exclude the small equality cases in the rank optimization and the two families of maximal even-dimensional orthogonal Grassmannians.

For distinct $x, y \in \mathbb{P}^1$, choose nonzero representatives of their evaluation functionals and let $t \neq 0$. Set

$$h_+ = \text{ev}_x + t \text{ev}_y, \quad h_- = \text{ev}_x - t \text{ev}_y, \quad \ell = \text{ev}_x^{(2n)} - t^2 \text{ev}_y^{(2n)}. \quad (5.2)$$

Rescaling the representatives only reparametrizes this family. For $\varepsilon \in \{+, -\}^L / \{\varepsilon \sim -\varepsilon\}$, let Σ_ε be the closure of the tuples satisfying $U_\nu \subset \ker h_{\varepsilon_\nu}$. The all-equal pattern is the common-secant variety Σ_L of Proposition 4.4. Define also

$$\mathfrak{F}_L = \{([\ell], (U_\nu)) : \text{rank } H_\ell = k, H_\ell|_{U_\nu \times U_\nu} = 0\}, \quad \Phi_L = \overline{\text{pr}_{X_L} \mathfrak{F}_L}. \quad (5.3)$$

The closure in this definition is essential: the generic annihilator of Φ_L will be nondegenerate, but its specializations need not be.

Theorem 5.1 (Component classification and determinantal scheme). *Assume (5.1).*

(i) *If $a < b$, the scheme $\mathcal{D}_L$ is integral and Cohen–Macaulay of codimension a , with support Φ_L . Its general point has multiplication corank one, a unique projective annihilator of Hankel rank k , and a smooth determinantal neighbourhood. The projection from $\mathfrak{F}_L$ is birational onto Φ_L .*

(ii) *If $a = b$, the scheme $\mathcal{D}_L$ is reduced and Cohen–Macaulay of pure codimension a . Its irreducible components are exactly*

$$\Phi_L \quad \text{and} \quad \Sigma_\varepsilon \quad (\varepsilon \in \{+, -\}^L / \{\pm 1\}).$$

There are $1 + 2^{L-1}$ components. The general point of each has multiplication corank one and is smooth in the determinantal scheme. The generic annihilator has rank k on Φ_L and rank two on every Σ_ε .

(iii) *If $b < a$, the maximal-dimensional irreducible components of $\Delta^{(L)}$ are exactly the 2^{L-1} varieties Σ_ε , of codimension b . Their general points have multiplication corank one and are smooth in $\mathcal{D}_L$. Every Σ_ε is rational.*

(iv) *In the expected-codimension range $a \leq b$, the fundamental cycle is*

$$[\mathcal{D}_L] = c_a(\mathcal{E}^*) = \left[\prod_{\nu=1}^L \prod_{1 \leq i \leq j \leq c} (1 + x_{\nu i} + x_{\nu j}) \right]_a \quad \text{in } A^a(X_L), \quad (5.4)$$

where $x_{\nu i}$ are the formal Chern roots of $\mathcal{S}_\nu^$ and the brackets select codimension a .*

Part (iii) classifies maximal-dimensional components, not possible smaller components or embedded structure in the excess range. Parts (i) and (ii), by contrast, concern the entire determinantal scheme. Formula (5.4) uses the classical determinantal cycle formula after the expected codimension and reducedness have been established; the cycle formula itself is not a new general intersection-theoretic principle.

5.1. The two extremal incidences.

Lemma 5.2 (Strict rank gap). *Under (5.1), every annihilator incidence stratum has dimension strictly less than $m_X - \min(a, b)$, except possibly the strata*

$$r = 2, \quad s_1 = \cdots = s_L = 1; \quad r = k, \quad s_1 = \cdots = s_L = c.$$

Their dimensions are respectively $m_X - b$ and $m_X - a$.

Proof. Use $D_{r,s}$ and h_r from Lemmas 4.1 and 4.2. If $r = k$, the radical is zero and $s = c$. Suppose $r < k$. For $s = 0$, $LD_{r,0} - h_r = (Lc - 2)r + 1 > b$. For $1 \leq s \leq c - 2$, the lower endpoint $r = 2s$ in the feasible interval gives

$$g(s) = 1 + s(Lc - 4) - Ls(s - 1)/2.$$

Its coefficient before minimizing in r is $L(c - s) - 2 \geq 0$. Moreover,

$$g(1) = b, \quad g(c - 2) - b = (c - 3)\{L(c + 2)/2 - 4\} > 0.$$

Concavity places every integer $2 \leq s \leq c - 2$ strictly above b . At $s = 1$ the coefficient $L(c - 1) - 2$ is strictly positive, so equality requires $r = 2$.

For $s = c - 1$ and $L = 1$, the calculation in Lemma 4.3 gives at least $a + k - 2c + 1 \geq a + 2$. For $s = c - 1$ and $L \geq 2$, it gives at least $b + (c - 2)\{L(c + 1)/2 - 4\} > b$. For $s = c$, $r < k$, it gives at least $a + 1$. Thus, for each fixed nonexceptional r and every feasible s , $LD_{r,s} - h_r > \min(a, b)$. Averaging this inequality over the L choices s_ν gives $\sum_\nu D_{r,s_\nu} - h_r > \min(a, b)$. At $r = 2$, replacing any $s_\nu = 1$ by zero increases this sum. The two displayed dimensions follow from $D_{2,1} = c$ and $D_{k,c} = q$. $\square$

The proof explains the two ends of the codimension formula: an arbitrary nondegenerate Hankel form imposes q isotropy equations per contact, whereas a rank-two form leaves a choice between two hyperplanes. Intermediate ranks incur a strict dimension loss in (5.1).

Lemma 5.3 (Signed secant incidences). *The 2^{L-1} varieties Σ_ε are distinct irreducible rational subvarieties of $\Delta^{(L)}$, each of codimension b . Their incidence parametrizations are birational. A general point of any one of them admits exactly one rank-two projective annihilator.*

Proof. A general point $[h]$ of the secant threefold S_n has a unique unordered secant decomposition. Indeed, evaluations at four distinct points are linearly independent for $n \geq 3$, by interpolation; the corresponding confluent statement follows by Hermite interpolation. Thus two different secant lines cannot meet away from their endpoints, and tangent or limiting cases form a proper closed subset. The parameter space is birational to a projective-line bundle over $\text{Sym}^2 \mathbb{P}^1 = \mathbb{P}^2$, hence is rational. Sending $[h_+]$ to $[h_-]$ in (5.2) is a rational involution, independent of the ordering of the two evaluation points.

Normalize the sign pattern by $\varepsilon_1 = +$. Over the dense open set of distinct secants with nonzero coefficients, the signed incidence is a product of relative Grassmannians, with dimension $3 + Lc(k - 1 - c) = m_X - b$. It is irreducible and rational: its underlying hyperplane bundles are trivial on a dense open subset of the rational base. The functional ℓ in (5.2) annihilates every product, for either sign, so its image closure lies in $\Delta^{(L)}$.

Here is the projection calculation explicitly. Fix $h \in S_n$ in the above open set. In $\text{Gr}(c, \ker h)$ the exceptional planes admitting another secant hyperplane are the projection of

$$\{(h', U) : h' \in S_n \setminus \{h\}, U \subset \ker h \cap \ker h'\}.$$

Its dimension is at most $3 + c(k - 2 - c)$, strictly smaller than $c(k - 1 - c)$ because $c > 3$; if $c > k - 2$ it is empty. Therefore a general first plane determines h uniquely. This proves generic injectivity and, in characteristic zero, birationality of the signed projection. The recovered h also determines its partner and the rank-two annihilator. For a general tuple no U_ν lies in both partner hyperplanes. Consequently two different normalized sign patterns have different images. The exact-rank-two stratum not represented by two distinct evaluations has smaller dimension; it cannot affect these generic assertions. $\square$

Lemma 5.4 (The nondegenerate incidence). *The variety $\mathfrak{F}_L$ is smooth and irreducible of dimension $m_X - a$. Its image contains a point whose first plane lies in no secant hyperplane.*

Proof. Over the open set of nonsingular Hankel forms, the fibres are products of $\text{OGr}(c, k)$. The restriction equations have independent differentials, as in Lemma 4.2. Since $2c < k$, the special orthogonal group acts transitively on these isotropic Grassmannians; a Witt basis proves transitivity, and the special orthogonal group over $\mathbb{C}$ is connected. Thus the fibres are smooth and geometrically irreducible of dimension $c(k - c) - q$. The base is a smooth irreducible open subset of $\mathbb{P}^{p-1}$, so the relative incidence is smooth and irreducible, with the stated dimension.

Fix one nonsingular form H and put $d_H = c(k - c) - q$. In any hyperplane K the restricted form has radical of dimension zero or one. In the nondegenerate case its isotropic c -planes have dimension $d_H - c$. In the one-dimensional radical case, the strata avoiding and containing the radical have dimensions respectively

$$d_H - c, \quad d_H - (k - c - 1) \leq d_H - c,$$

when nonempty. These formulas follow by the same choice-of-radical-intersection and lift calculation as Lemma 4.2. Varying K among secant hyperplanes contributes only three parameters. Their union in $\text{OGr}(c, k)$ therefore has dimension at most $d_H - c + 3 < d_H$. Choose U_1 outside it and choose the other isotropic planes arbitrarily. This proves the last assertion. $\square$

5.2. Corank one and the residual multiplication map. Let $\mathfrak{J} \subset \mathbb{P}(V_{2n}^*) \times X_L$ be the scheme defined by $\ell\mu = 0$.

Lemma 5.5 (Corank-one chart). *Above the open subset on which μ has rank at least $p - 1$, the projection from $\mathfrak{J}$ to $\mathcal{D}_L$ is an isomorphism. At a point of corank one, infinitesimal pairs satisfy*

$$\dot{\ell}(uv) + H_\ell(\dot{u}, v) + H_\ell(u, \dot{v}) = 0 \quad (u, v \in U_\nu, 1 \leq \nu \leq L). \quad (5.5)$$

Here $\dot{U}_\nu \in \text{Hom}(U_\nu, V_n/U_\nu)$ and $\dot{\ell}$ is taken modulo $\mathbb{C}\ell$.

Proof. On a nonzero $(p - 1)$ -minor, invertible row and column operations reduce μ to an identity block of size $p - 1$ and a remaining row of length $e - p + 1$. Its entries generate the maximal-minor ideal. The normalized last coordinate of a projective left kernel vector is one; the other coordinates are uniquely eliminated by the identity block. The remaining equations are exactly that row. This proves the scheme isomorphism, not only a bijection of points. Differentiating the product equations gives (5.5). $\square$

Lemma 5.6 (Generic smoothness of the signed components). *If $b \leq a$, a general point of each Σ_ε has corank one and the scheme $\mathcal{D}_L$ is smooth there of codimension b .*

Proof. The strict rank gap excludes every intermediate-rank annihilator at a general point of Σ_ε . If $b < a$, the full-rank incidence has smaller dimension and is excluded as well. If $a = b$ and full-rank annihilators occurred over a dense subset of Σ_ε , their fibres would have positive

dimension: the projective annihilator space already contains its rank-two point, and its full-rank part is open. This would force $\dim \mathfrak{F}_L > m_X - a$, a contradiction. Lemma 5.3 then implies corank one. Indeed, any further annihilator would give a projective line of annihilators, and hence either a forbidden rank or more than one rank-two point.

For the tangent calculation take the distinct evaluation points in (5.2). Let g be the binary quadratic vanishing at them. Then

$$E = \text{rad } H_\ell = gV_{n-2}, \quad U_\nu = W_\nu \oplus \mathbb{C}a_\nu, \quad W_\nu = U_\nu \cap E = gK_\nu,$$

where the K_ν are independent general $(c-1)$ -planes in V_{n-2} . The inequality determining this regime is precisely

$$a - b = L \binom{c}{2} - (2k - 5) \geq 0. \quad (5.6)$$

Apply Theorem 1.1, already proved in Section 4, with $k' = k - 2$ and $c' = c - 1$. Its hypotheses hold and (5.6) is its saturation condition. Consequently, on a nonempty open subset of the signed incidence,

$$\sum_\nu W_\nu^2 = g^2 V_{2n-4}, \quad \dim \sum_\nu W_\nu^2 = p - 4. \quad (5.7)$$

There is no inductive use of the present structural theorem.

Equation (5.5) on $W_\nu \times W_\nu$ first requires $\dot{\ell}$ to annihilate (5.7). Thus $\dot{\ell}$ has dimension $(p-1) - (p-4) = 3$. For each such $\dot{\ell}$, the remaining equations consist of the $c-1$ pairs (w, a_ν) and the pair (a_ν, a_ν) . Choose $b_\nu \in V_n$ with $H_\ell(a_\nu, b_\nu) = 1$. The b_ν -coordinates of $\dot{U}_\nu(w)$ and $\dot{U}_\nu(a_\nu)$ solve these c equations independently; the last coefficient is two, which is nonzero. Thus the tangent dimension of $\mathfrak{I}$, and by Lemma 5.5 of $\mathcal{D}_L$, is

$$3 + \sum_\nu \{c(k-c) - c\} = m_X - b.$$

The local dimension is at least this number because the point lies on Σ_ε . Equality of dimension and tangent dimension proves regularity of the local ring and the stated smoothness. $\square$

Identity (5.6) is a deformation-theoretic interpretation of the change of regime. Saturation of the multiplication map after removing the two evaluation points is exactly the condition under which the rank-two component acquires its predicted smooth normal space. This is stronger than locating a minimum in an incidence count.

5.3. The whole scheme in expected codimension.

Proof of Theorem 5.1. Suppose first that $a \leq b$. On a product of standard affine Grassmannian charts the coordinate ring is a polynomial ring in m_X variables over $\mathbb{C}$, the matrix of μ has size $p \times e$, and $I = I_p(\mu)$ is its maximal-minor ideal. On every chart meeting the failure scheme, Theorem 1.1 gives height $I \geq a$ and the maximal-minor height inequality gives height $I \leq e - p + 1 = a$. Since the ring is regular, grade $I = a$. The Eagon–Northcott grade criterion therefore provides a free resolution of R/I of length a ; in particular it is Cohen–Macaulay and unmixed of codimension a [20, 21]. This application uses the grade criterion, not a claim that the entries of μ are independent variables.

Every component consequently has dimension $m_X - a$. By Lemma 5.2, its generic annihilator must have rank k , or, when $a = b$, rank two. In the first case it is dominated by the irreducible incidence $\mathfrak{F}_L$, so there is at most one such component. Lemma 5.4 supplies a point outside every signed secant variety; purity and the strict rank gap force a full-rank component through the resulting non-secant part of the scheme. Hence this component exists and is Φ_L , with dimension $m_X - a$.

The full-rank incidence has this same dimension. Its generic fibre is therefore zero-dimensional, but this fibre is the full-rank open subset of the projective annihilator space. It follows that the generic multiplication corank is one. The generic fibre is a single point, and Lemma 5.5 identifies a dense open subset of Φ_L with an open subset of the smooth variety $\mathfrak{F}_L$. This proves birationality and generic scheme smoothness there.

If $a < b$, the rank-two incidence has smaller dimension, leaving exactly Φ_L . If $a = b$, every distinct Σ_ε has the dimension of a component, and Lemma 5.3 identifies all the rank-two components; the confluent rank-two base is a proper lower-dimensional subset. There are no others by the strict rank gap. Lemma 5.6 proves generic scheme smoothness on them. A Cohen–Macaulay scheme has no embedded associated primes; if it is generically reduced at every minimal prime, it is reduced. Applying this fact proves (i) and (ii), including integrality in (i).

If $b < a$, all maximal-dimensional components have dimension $m_X - b$. The full-rank incidence and all the intermediate-rank incidences are too small to dominate one. Lemma 5.3 therefore gives exactly the listed components, their codimension and rationality, and Lemma 5.6 gives their generic corank and smoothness. This proves (iii).

Finally apply the Thom–Porteous formula to the dual map $V_{2n}^* \otimes \mathcal{O} \longrightarrow \mathcal{E}^*$, with source rank p , target rank e , and rank bound $p - 1$ [22, 23]. The expected codimension is a , as verified above, so the formula is $c_a(\mathcal{E}^*)$. The roots of $\text{Sym}^2 \mathcal{S}_\nu^*$ are $x_{\nu i} + x_{\nu j}$, $i \leq j$, which gives (5.4). Reducedness shows that in the equality case this is the sum of the component cycles with coefficient one. No excess-range use of the expected-codimension formula is made. $\square$

5.4. Real native strata and local conditioning.

Corollary 5.7 (All signed components have positive native points). *For a native interpolation matrix J , every Σ_ε has a Zariski-dense set of real points admitting positive, globally separated native realizations. If (5.1) holds and $b \leq a$, some such points are smooth points of $\mathcal{D}_L$ of real local codimension b . Their pullbacks to loading space have the same local codimension.*

Proof. Choose real $x, y > D$ and sufficiently small $t > 0$. Every coordinate of $-J \text{ev}_x$ is positive, so both $-Jh_+$ and $-Jh_-$ are strictly positive. Independently for each contact, choose a general $(k - c)$ -plane T_ν containing $-Jh_{\varepsilon_\nu}$ and avoiding the vectors obtained from it by nonconstant coordinate sign changes. The fixed-space construction in Theorem 6.1 then gives the required globally separated positive loading and $U_\nu \subset \ker h_{\varepsilon_\nu}$. These choices contain a real open subset of the signed incidence. Its birational parametrization is defined over $\mathbb{R}$ and is an isomorphism on dense open subsets. A nonzero real polynomial cannot vanish on this real open subset, so it meets the generic corank-one and residual-spanning opens of Lemma 5.6. The real smooth codimension equals the complex codimension there. The loading submersions of Theorem 6.1 transfer the assertion. In the finite sign test, s and $-s$ define the same unordered coordinate partition. Positivity specifies a Euclidean-open, Zariski-dense real image, not a positivity-defined Zariski-open set. $\square$

Corollary 5.8 (Local small-singular-value exponent). *At any real smooth corank-one point furnished by Corollary 5.7, there is a loading coordinate neighbourhood, disjoint from the other components, on which the least multiplication singular value γ_{mult} satisfies*

$$C_1 \text{dist}(A, |\mathcal{D}_L|_{\text{load}}) \leq \gamma_{\text{mult}}(A) \leq C_2 \text{dist}(A, |\mathcal{D}_L|_{\text{load}}). \quad (5.8)$$

Here the neighbourhood has compact closure in a fixed frame chart and in a fixed full-rank loading sign chamber. On a sufficiently small coordinate product box, a probability law with density bounded above and below by positive constants satisfies

$$c_1 \epsilon^b \leq \Pr\{\gamma_{\text{mult}} \leq \epsilon\} \leq c_2 \epsilon^b \quad (0 < \epsilon < \epsilon_0). \quad (5.9)$$

At a real smooth corank-one point of Φ_L in the range $a \leq b$, the same assertion holds with a in place of b .

Proof. Use the Schur-complement row of Lemma 5.5. Bounded invertible matrix operations compare its Euclidean norm with γ_{mult} . At a smooth scheme point of codimension b , the differentials of its entries span a conormal space of dimension b . Select b independent entries as normal coordinates. The whole row vanishes on the failure submanifold; bounded derivatives give the upper distance estimate, and the selected coordinates give the lower estimate. The loading submersion and compact chart bounds preserve these comparisons. Integration in the b normal coordinates, with bounded Jacobian and density on the product box, gives (5.9). The full-rank case is identical with codimension a . $\square$

These are local probability laws near specified smooth strata. They do not assert a global tail for arbitrary random loadings or constants uniform in c, k, L . In particular, singular intersections and smaller components in the excess range are not covered by this local tube calculation.

6. NATIVE REALIZATION AND GLOBAL SEPARATION

All spaces in this section are real. The following result holds for any fixed $J \in \text{GL}_k(\mathbb{R})$, not just the interpolation matrix.

Theorem 6.1 (Exact fixed-space realization). *Let $1 \leq d < k$, $c = k - d$, $U \in \text{Gr}(c, k)$, and $T = JU^\perp$. The space U is the normal-polynomial space of a local immersed image q_A at $q_A(e_1) > 0$ if and only if*

$$T \cap \mathbb{R}_{>0}^k \neq \emptyset. \quad (6.1)$$

There is such a realization with

$$q_A^{-1}(q_A(e_1)) = \{e_1, -e_1\} \quad (6.2)$$

if and only if, in addition,

$$D_s T \neq T \quad \text{for every } s \in \{\pm 1\}^k \setminus \{\mathbf{1}, -\mathbf{1}\}. \quad (6.3)$$

Here $D_s = \text{diag}(s)$. On each sign chamber of the full-rank loading domain with Ae_1 nowhere zero, the map

$$A \mapsto U(A) = J^\top (\text{ran}(\text{diag}(Ae_1)A))^\perp$$

is a submersion onto its positive realizable image. That image is nonempty, Euclidean open, and Zariski dense.

Proof. Put $z = Ae_1$. The tangent space is $T = \text{ran}(\text{diag}(z)A)$ and contains $t = z^2 > 0$. Taking orthogonal complements in $U = J^\top T^\perp$ gives $T = JU^\perp$ and proves necessity of (6.1). Conversely choose $t > 0$ in T , extend it to a basis $t, t_2, \dots, t_d$, and set

$$z_i = \sqrt{t_i}, \quad A = [z, \text{diag}(z)^{-1}t_2, \dots, \text{diag}(z)^{-1}t_d]. \quad (6.4)$$

Then $\text{diag}(z)A$ has this prescribed basis, and $q_A(e_1) = t$.

Equality of coordinatewise squares is equivalent to $A\eta = D_s z$ for a sign vector s . Because D_s commutes with $\text{diag}(z)$,

$$D_s z \in \text{ran } A \iff D_s t \in T. \quad (6.5)$$

If $D_s T = T$ for a nonconstant s , the right side holds for every $t \in T$; positivity ensures $D_s t \neq \pm t$, giving another preimage. Conversely, under (6.3), each $T \cap D_s T$ is a proper linear subspace of T . Their finite union cannot cover the relatively open positive cone in (6.1). Choose t outside that union and use (6.4). Equation (6.5) leaves only the constant signs, hence exactly the two preimages in (6.2).

For the submersion statement, on a fixed sign chamber the change of frames $A \mapsto B = \text{diag}(z)A$ is a diffeomorphism onto full-rank frames whose first column is strictly positive. Its inverse has

$z_i = \pm\sqrt{B_{i1}}$ with the fixed signs and $A_{ij} = B_{ij}/z_i$. The ordinary frame-to-plane map is a submersion. Orthogonal complement and multiplication by fixed J preserve that property. Positivity persists under small changes of a frame, and planes containing $\mathbf{1}$ show nonemptiness. A polynomial vanishing on a nonempty real open subset of an affine Grassmannian chart vanishes identically, proving Zariski density. $\square$

The obstruction in (6.3) has a concrete interpretation. If I is the set of positive signs of s , then $D_s T = T$ precisely when

$$T = (T \cap \mathbb{R}^I) \oplus (T \cap \mathbb{R}^{I^c}).$$

Thus a prescribed normal space may be locally realizable yet admit no globally separated realization at that same space. Theorem 6.1 classifies exactly this distinction.

Lemma 6.2 (Generic global distance). *A nonempty Zariski-open subset of the full-rank loading domain with Ae_1 nowhere zero satisfies (6.2). For each such fixed A , the global image has one embedded analytic germ at $b_0 = q_A(e_1)$. For every compact set $\mathcal{R} \subset \text{Sym}_k^{>0}$ there is a neighbourhood of b_0 on which the distance to that germ equals the global distance, and the minimizing projection is analytic with uniform bounds for $R \in \mathcal{R}$.*

Proof. For a nonconstant sign vector, the condition $D_s Ae_1 \in \text{ran } A$ is algebraic, defined by augmented $(d+1)$ -minors. It is proper: choose a vector z with no zero entries and a d -plane containing z but not $D_s z$, then take a frame with first column z . Removing the finitely many such proper sets proves the first assertion. The ambient loading space is the irreducible affine space $\mathbb{R}^{kd}$; full rank and nonzero first coordinates specify a Zariski-open domain, so these finite open intersections remain dense.

Full rank gives

$$\|q_A(\eta)\|_2 \geq k^{-1/2} \|A\eta\|_2^2 \geq k^{-1/2} \sigma_{\min}(A)^2 \|\eta\|_2^2,$$

so q_A is proper. Its derivative is injective at both preimages, and their local image germs coincide by evenness. Properness excludes a remote sheet accumulating at b_0 . The trial image b_0 bounds the distance by $C\|b - b_0\|^2$. Uniform positive metric bounds on $\mathcal{R}$ therefore force all minimizing images into this germ when b is close enough to b_0 . The positive tangent Hessian and the implicit function theorem give the projection tube. Its radius and derivative constants depend on A , the local germ and $\mathcal{R}$; no uniformity over all generic loadings is claimed. $\square$

Proof of Theorem 1.2. Necessity is the dimension bound in Section 2. If it holds and $d \geq 2$, then $c = k - d \geq 5$. Theorem 1.1 with $L = 1$ gives a nonempty complex Zariski-open full-product locus. Its defining minors have real coefficients and some minor is a nonzero polynomial on a real affine chart. A nonzero real polynomial cannot vanish on any nonempty real open set. Thus the full-product locus meets the positive realizable image of Theorem 6.1.

On pivot charts normal frames are rational functions of A , so full product rank pulls back to a nonempty Zariski-open subset of the irreducible loading domain. Intersect it with Lemma 6.2's dense open set. Theorem 2.2 gives recovery from the global distance for every $\theta \in \Omega$ with the same J . Writing $k = d + c$ gives $c^2 - 3c - 4d + 2 \geq 0$, whose positive root yields (1.3). Formula (2.1) gives the unrestricted ambiguity. $\square$

For $d = 2, 3, 4, 5, 6$, the optimal k are 7, 8, 10, 11, 13. At the optimum the full symmetric measurement $S \mapsto (a_i^\top S a_i)_i$ is necessarily noninjective for $d \geq 5$: check $d = 5, 6$ directly, and use $k_{\min} \leq 2d < d(d+1)/2$ for $d \geq 7$. This is not full-symmetric phase retrieval in disguise.

Corollary 6.3 (Independent native designs and the real secant family). *Fix $4 \leq c < k$ and a native interpolation matrix J . For L independently chosen loading matrices of size $k \times (k - c)$, generic globally separated contacts recover the common native moment parameter if and only if $Lq \geq p$.*

The common-secant family Σ_L contains real, positive, globally separated native realizations. At its general such points its loading pullback has real local codimension $Lc - 3$.

Proof. Apply the product of the loading submersions and generic global-separation opens to Theorem 1.1, using real nonvanishing of minors as above. For the second assertion choose distinct $x, y > D$ and $h = \text{ev}_x + \varepsilon \text{ev}_y$ with $\varepsilon > 0$. By (3.1), every coordinate of $Jw_n(t)$ is negative for $t > D$. Thus $t = -Jh > 0$. A general d -plane T containing t avoids each of the finitely many vectors $D_s t$ for nonconstant s , since none is proportional to t and $d < k$. Such a $T = JU^\perp$ has $U \subset \ker h$ and, by (6.4)–(6.5), a globally separated realization. Choose the T_ν independently. These choices form real open subsets in the secant incidence. Its generically one-to-one projection is birational in characteristic zero and is an isomorphism on dense open subsets. The loading submersion consequently preserves its codimension there. $\square$

The exact formula (1.2) is an algebraic statement over $\mathbb{C}$. The corollary supplies a real native component of that dimension when the secant term is minimal. It does not infer real points on every complex component in the other regime. It also does not replace the independent-design assumption by multiple arbitrary points on one fixed quadratic image.

7. RANK STRATA AND QUANTITATIVE INVERSION

Let Δ_r be the determinantal scheme on $\text{Gr}(c, V_n)$ where multiplication has rank at most r . At a point of rank *exactly* r , put $K_U = \ker \mu_U$ and $Q_U = V_{2n}/U^2$.

Proposition 7.1 (Infinitesimal rank map). *The tangent space to Δ_r is the kernel of*

$$\beta_U : \text{Hom}(U, V_n/U) \longrightarrow \text{Hom}(K_U, Q_U),$$

$$\beta_U(\phi) \left(\sum_i f_i \odot g_i \right) = \sum_i \{ \phi(f_i)g_i + f_i\phi(g_i) \} \pmod{U^2}. \quad (7.1)$$

If β_U is onto, the scheme is smooth there of codimension $(q-r)(p-r)$. The corresponding native contact fibre has dimension $p-r$.

Proof. Express nearby planes as graphs and differentiate products. A different lift of ϕ changes the result only by an element of U^2 . Near a nonzero r -minor, invertible block operations reduce multiplication to an invertible r -block and a Schur complement. Rank at most r is equivalent to the vanishing of that complement, whose differential is (7.1). Its kernel is the scheme tangent space, and surjectivity gives smoothness with the displayed codimension. The fibre assertion is Theorem 2.2. $\square$

The same construction on X_L uses the direct sum of the graph derivatives. The assertion is about the scheme defined by minors, and does not presuppose its reducedness.

Use coefficient Euclidean norms, the induced symmetric tensor norm on multiplication, direct-sum Frobenius norms on blocks, and the standard Grassmannian Riemannian distance. Let $\Delta_{\mathbb{R}}$ denote the real failure set. Let γ be the least singular value of an injective $\mathcal{L}$.

Theorem 7.2 (Stability and transverse degeneracy). *Suppose $m_0 I \leq B_\nu \leq M_0 I$ and $\hat{G}_\nu = G_\nu + E_\nu$, with $\max_\nu \|E_\nu\|_{\text{op}} \leq (2M_0)^{-1}$. Then*

$$\left\| \mathcal{L}^\dagger(\hat{G}_\nu^{-1})_\nu - \theta \right\|_2 \leq \frac{2M_0^2}{\gamma} \left(\sum_\nu \|E_\nu\|_F^2 \right)^{1/2}. \quad (7.2)$$

The least multiplication singular value is bounded above by a constant times distance to $\Delta_{\mathbb{R}}$. Near a real corank-one point at which the corresponding kernel-to-cokernel differential is onto, it is bounded above and below by positive local constants times that distance. The same local comparison holds in loading space for distance to the pulled-back failure set.

Proof. The inverse identity gives $\hat{G}_\nu^{-1} - B_\nu = -\hat{G}_\nu^{-1}E_\nu B_\nu$. The assumed bounds make its Frobenius norm at most $2M_0^2\|E_\nu\|_F$. Apply $\mathcal{L}^\dagger$, whose norm is γ^{-1} .

In finitely many compact orthonormal frame charts, multiplication is Lipschitz in the plane coordinates. Singular values are Lipschitz in matrix entries and vanish on the real failure set, giving the upper estimate. At corank one, the Schur complement is a row of length $Lq - p + 1$. Bounded invertible block operations compare the least singular value with the norm of this row. Surjectivity of its differential gives local coordinates in which the row is a coordinate projection, proving the lower estimate as well. For fixed J , passage from an orthonormal polynomial frame to $J^\top Z$ changes norms by bounded positive factors. The loading submersion gives local product coordinates and transfers the distance comparison. $\square$

No linear distance law is asserted at nontransverse or singular points. Compact design sets disjoint from failure have a positive lower stability constant by continuity; this alone is not a dimension-uniform numerical estimate or a random-loading tail bound.

8. STRUCTURED COMPRESSION BEYOND HANKEL MATRICES

Let $\mathcal{K} \subset \text{Sym}_k$ have dimension s and contain a positive matrix. Write $\rho(\mathcal{K}, c)$ for the maximum rank of $X \mapsto X|_{N \times N}$ over $N \in \text{Gr}(c, k)$.

Proposition 8.1 (Generic transfer). *For generic full-rank native loadings with Ae_1 nowhere zero, compression has rank $\rho(\mathcal{K}, c)$ and the global contact fibre in an open positive subset of $\mathcal{K}$ has dimension $s - \rho(\mathcal{K}, c)$.*

Proof. Maximal compression rank is a nonempty Zariski-open condition. Apply Theorem 6.1 with $J = I$ and Lemma 6.2. Positive block inversion preserves fibres, and rank-nullity gives the dimension. $\square$

This is a transfer principle, not a computation of ρ for an unspecified family. The following computation includes multiplicities in a fixed spectral decomposition.

Theorem 8.2 (Block-spectral rank). *Let $P_1, \dots, P_s$ be nonzero mutually orthogonal real symmetric projections with sum I_k , and let $\mathcal{K} = \text{span}\{P_1, \dots, P_s\}$. Then*

$$\rho(\mathcal{K}, c) = \min\{s, \binom{c+1}{2}\}.$$

For the diagonal family the sharp threshold is $\binom{c+1}{2} \geq k$, with least ambient dimension $d + \lceil (1 + \sqrt{1 + 8d})/2 \rceil$ at fixed $d = k - c$. For the family of real symmetric circulant matrices it is $\binom{c+1}{2} \geq \lfloor k/2 \rfloor + 1$.

Proof. The upper bound is dimensional. Suppose $c \leq s$. Choose one unit vector in each $\text{ran } P_i$. In their span, a normal frame with rows $u_i^\top$ gives compressed matrices $u_i u_i^\top$. The c vectors e_a followed by the $c(c-1)/2$ vectors $e_a + e_b$ have linearly independent outer products: first inspect off-diagonal entries, then diagonal entries. Select the first s when $s \leq q$, or retain all q and add rows when $s > q$. Starting with the e_a gives full column rank. Column-orthonormalization transforms every compressed matrix by the same invertible congruence and therefore preserves linear independence. If $c > s$, take a c -plane containing one unit vector from every $\text{ran } P_i$ and extend their span. Evaluation on these distinguished vectors separates the s coefficients. This proves the rank formula in all cases.

For diagonal matrices, $s = k$ and substitution $k = d + c$ gives $c^2 - c - 2d \geq 0$. A real symmetric circulant matrix is diagonal in the real Fourier decomposition: the zero frequency, the Nyquist frequency when present, and the paired sine-cosine frequencies give $\lfloor k/2 \rfloor + 1$ orthogonal spectral projections. Its eigenvalue is arbitrary on each such block, so the preceding formula applies. $\square$

Independent Bernoulli log-odds coordinates with information weights λ_i have inverse information $\text{diag}((\lambda_i p_i(1 - p_i))^{-1})$, providing a genuine positive diagonal information family. The block-spectral theorem concerns the stated matrix families and does not claim that every such family is generated by the calibrated polynomial experiment, or compute ranks for all Toeplitz or graph-sparse families.

9. THE PHYSICAL BOUNDARY EXPERIMENT

The algebraic extension through $v = 0$ in Section 3 is useful for differentiation, but it does not remove the physical restriction $v \geq 0$. We now identify the actual local experiment. The argument is the finite-cell likelihood expansion underlying constrained Gaussian limits [7, 16]; all normalizations and the synchronized profile are specified here.

Let $u \in \mathbb{R}^{k+1}$ denote the nuisance log-odds and root-sum coordinates. Observe independent categorical samples with $n_j/N \rightarrow \lambda_j > 0$, and let $\Delta_N = (\Delta_{N,u}, \Delta_{N,v})$ be the score at $(u, v) = (0, 0)$ divided by $\sqrt{N}$. Partition

$$I = V^\top \text{diag}(\lambda_j \kappa_j) V = \begin{pmatrix} I_{uu} & I_{uv} \\ I_{vu} & I_{vv} \end{pmatrix}, \quad Q = I_{vv} - I_{vu} I_{uu}^{-1} I_{uv} = R^{-1}.$$

Theorem 9.1 (Cone likelihood and synchronized profile). *Fix a compact set $K \subset \mathbb{R}^{k+1} \times \mathbb{R}_{\geq 0}^k$, with the Euclidean product norm. For $(a, h) \in K$, the physical local alternatives $u = a/\sqrt{N}$, $v = h/\sqrt{N}$ satisfy, uniformly on compact index sets,*

$$\log \frac{dP_{N,a,h}}{dP_{N,0,0}} = (a, h)^\top \Delta_N - \frac{1}{2}(a, h)^\top I(a, h) + o_P(1). \quad (9.1)$$

Here uniformity means that the supremum over K of the absolute remainder converges to zero in $P_{N,0,0}$ -probability. Under the corresponding alternatives,

$$Y_N = R(\Delta_{N,v} - I_{vu} I_{uu}^{-1} \Delta_{N,u}) \implies N(h, R). \quad (9.2)$$

After quotienting out unrestricted nuisance translation, this is a Gaussian shift with parameter restricted to the physical cone. For synchronized alternatives $h = q_A(\zeta)$, the profiled limiting log likelihood, relative to its value at $h = 0$, is

$$\frac{1}{2} Y^\top R^{-1} Y - \frac{1}{2} C_R(Y), \quad C_R(Y) = \inf_{\zeta} (Y - q_A(\zeta))^\top R^{-1} (Y - q_A(\zeta)). \quad (9.3)$$

The common physical root displacement has scale $N^{-1/4} \zeta$.

Proof. The finite set of clock probabilities is strictly positive on a two-sided neighbourhood of the baseline. Its derivatives through order three are bounded on a smaller neighbourhood. At $u = a/\sqrt{N}$, $v = h/\sqrt{N}$, expand each cell log probability. Normalization makes the mean linear term vanish, the expectation of the Hessian is minus the score covariance, and the summed cubic remainder is $O(N^{-1/2})$ on compact index sets. The centred Hessian fluctuation is $o_P(1)$. The finite-dimensional independent-group central limit theorem gives $\Delta_N \Rightarrow N(0, I)$ at the baseline and proves (9.1). Under the local alternatives its mean tends to $I(a, h)$ and its covariance to I , either by the same cell calculation or the likelihood change of measure. The efficient score in (9.2) has mean Qh and covariance Q , proving the assertion after multiplication by R .

In the limiting score experiment the nuisance block is independent of the efficient block. Its mean is $I_{uu}a + I_{uv}h$, which is an unrestricted translation as a varies. Discarding this nuisance translation yields (9.2); this is a quotient, not a claim that an unknown nuisance parameter can be simulated without information. Profiling the efficient Gaussian likelihood over $h = q_A(\zeta)$ and completing the square gives (9.3). The map q_A is proper for full-rank A , so the infimum is attained and its optimizing ζ are bounded whenever Y is bounded. Uniform LAN therefore

also gives the iterated compact-localization limit of profiled likelihoods: first restrict (a, ζ) to a fixed compact set, pass to the limit, and then exhaust the limiting experiment by such sets. No assertion about remote maximizers of an unrestricted finite-sample likelihood is required. Finally $q_A(N^{-1/4}\zeta) = N^{-1/2}q_A(\zeta)$, proving the displacement scale. $\square$

The unrestricted vector Y is a statistic in the limiting experiment, not the constrained maximum likelihood estimator. The latter is a metric projection onto $\mathbb{R}_{\geq 0}^k$, or onto $q_A(\mathbb{R}^d)$ in the synchronized experiment. At the boundary it need not be normal; for the nonconvex synchronized image the projection can be set-valued. The distance value in (9.3) is nevertheless unambiguous.

Corollary 9.2 (Sharp root-amplitude scale on a physical ray). *Fix $a_0 \in \mathbb{R}^d \setminus \{0\}$ and $v_0 = q_A(a_0) \neq 0$. On the one-sided submodel with physical displacement ta_0 , $0 \leq t \leq TN^{-1/4}$, and nuisance $u = a/\sqrt{N}$ in a fixed bounded set, the minimax squared risk for estimating t is of order $N^{-1/2}$, for each fixed $T > 0$. Thus the amplitude scale is $N^{-1/4}$, whereas its squared amplitude is on the $N^{-1/2}$ parameter scale.*

Proof. The variance parameter is $v = t^2 v_0$. Set $J_0 = v_0^\top Q v_0 > 0$ and use the baseline efficient score to define

$$\hat{b}_N = \frac{v_0^\top Q Y_N}{\sqrt{N} J_0}, \quad \hat{t}_N = \sqrt{\max\{\hat{b}_N, 0\}}.$$

For completeness, let s_{je} be the baseline score of cell e in group j . There are finitely many such bounded vectors. Taylor expansion of the cell probabilities, rather than convergence in distribution, gives

$$\sup_{(a,h) \in K} \|\mathbb{E}_{a,h} \Delta_N - I_N(a,h)\| \leq C_K N^{-1/2}, \quad \sup_{(a,h) \in K} \|\text{Var}_{a,h} \Delta_N\|_{\text{op}} \leq C_K,$$

where I_N uses the actual allocations n_j/N and $I_N \rightarrow I$. The first bound sums the bounded second-order probability remainders; the second sums variances of independent bounded scores. The efficient linear map consequently has mean $h + o(1)$ and bounded covariance, uniformly on K . With $h = \sqrt{N} t^2 v_0$, the definition of $\hat{b}_N$ now gives bias $o(N^{-1/2})$ and variance $O(N^{-1})$, and hence $\mathbb{E}(\hat{b}_N - t^2)^2 \leq C/N$. This argument supplies the moment control separately from LAN. For $b \geq 0$, $(\sqrt{\max\{x, 0\}} - \sqrt{b})^2 \leq |x - b|$. Cauchy–Schwarz therefore gives the upper bound $C'/\sqrt{N}$.

For the lower bound take the two physical alternatives $t = 0$ and $t = \tau N^{-1/4}$ with the same nuisance, where $0 < \tau \leq T$ is sufficiently small. Their likelihood divergence is bounded by a constant times τ^4 , by (9.1) or its finite-cell Taylor proof. Their total variation is consequently bounded away from one. The two-point testing bound for squared loss gives a positive constant times the squared separation $\tau^2 N^{-1/2}$. The claim concerns this labelled nonnegative ray; it does not assert identifiability of the sign of a general loading vector. $\square$

10. DIRECT COUNT-TO-CONTACT EXPERIMENT COMPARISON

Here the observations differ from fixed-count categorical marks. Let the mark laws, hence κ_j , be known, and observe independent exposures

$$N_j \sim \text{Pois}(N\lambda_j), \quad j = 1, \dots, m.$$

This is the known-mark-law subexperiment of an observed-exposure protocol. Fix an interior centre $\lambda^0 > 0$. Write D for the derivative of the native moment functional at that centre:

$$D_{\cdot j} = -\frac{\rho_j}{(\lambda_j^0)^2 \kappa_j} w_{2n}(T_j), \quad \Lambda = \text{diag}(\lambda_j^0), \quad \Sigma = D\Lambda D^\top > 0.$$

The positivity follows from Vandermonde rank. For a local moment index $h \in \mathbb{R}^p$ in a fixed compact ball, define the least-favourable target submodel

$$\lambda_N(h) = \lambda^0 + N^{-1/2} B_* h, \quad B_* = \Lambda D^\top \Sigma^{-1}. \quad (10.1)$$

It has $\theta(\lambda_N(h)) = \theta(\lambda^0) + h/\sqrt{N} + O(N^{-1})$. Its local information is Σ^{-1} , since $DB_* = I_p$ and $B_*^\top \Lambda^{-1} B_* = \Sigma^{-1}$.

Let $\mathcal{L}$ be the known normal-compression operator, possibly rank deficient, and put $r = \text{rank } \mathcal{L}$. Choose an orthonormal coordinate frame F for its range. Generate independent $U_j \sim \text{Unif}(-1/2, 1/2)$, independently of the counts, and retain only

$$T_N = F^\top \mathcal{L} D \frac{N \cdot - N\lambda^0 + U}{\sqrt{N}}. \quad (10.2)$$

The randomization law is part of the protocol; its realized auxiliary values are not retained. This statistic is formed directly by fixed linear weights on the counts. No estimator of θ is constructed first. The small randomization prevents lattice encoding in this protocol: without such randomization or a genuine finite-precision coarsening, an exact real-valued linear encoding of an integer vector can be injective even when its real matrix has deficient rank. Weak convergence alone would not exclude that lattice encoding.

For experiments $\mathcal{A} = (P_h)$ and $\mathcal{B} = (Q_h)$ on the same parameter set, use

$$\delta(\mathcal{A}, \mathcal{B}) = \inf_K \sup_h \|K P_h - Q_h\|_{\text{TV}}, \quad \Delta(\mathcal{A}, \mathcal{B}) = \max\{\delta(\mathcal{A}, \mathcal{B}), \delta(\mathcal{B}, \mathcal{A})\}.$$

The infimum is over parameter-independent Markov kernels and $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$, equal to one half the L^1 distance for densities. All comparisons below use a fixed compact local parameter set.

Lemma 10.1 (Quantitative jittered Poisson comparison). *Uniformly for h in a fixed compact set, the distribution of*

$$X_N^\circ = (N \cdot - N\lambda^0 + U)/\sqrt{N}$$

under (10.1) is within $CN^{-1/5}$ in total variation of $N(B_ h, \Lambda)$. The constant depends on the fixed dimension, positive exposure bounds, and the compact index set. The full jittered vector and the raw counts are exactly equivalent experiments, by independent jittering and rounding. The raw local target experiment is therefore asymptotically equivalent to $Z \sim N(h, \Sigma)$.*

Proof. We give the density estimate, including the lattice interpolation and tails. In one coordinate write $K \sim \text{Pois}(N\lambda)$, with $0 < \lambda_- \leq \lambda \leq \lambda_+$, and centre first at its own mean. The density $f_{N,\lambda}$ of $(K - N\lambda + U)/\sqrt{N}$ is constant, equal to $\sqrt{N} \Pr(K = j)$, on each cell of width $N^{-1/2}$. Put $z_j = (j - N\lambda)/\sqrt{N}$ and let φ_λ be the centred normal density with variance λ . Stirling's formula with remainder

$$\log j! = (j + \tfrac{1}{2}) \log j - j + \tfrac{1}{2} \log(2\pi) + r_j, \quad 0 < r_j < 1/(12j),$$

and the Taylor formula for $\log(1 + u)$ give, uniformly for $|z_j| \leq 2N^{1/10}$ and all sufficiently large N ,

$$\left| \log \frac{\sqrt{N} \Pr(K = j)}{\varphi_\lambda(z_j)} \right| \leq C \{N^{-1/2}(1 + |z_j|^3) + N^{-1}(1 + |z_j|^4)\}.$$

Indeed $j = N\lambda(1 + z_j/(\sqrt{N}\lambda))$, the quadratic term is $-z_j^2/(2\lambda)$, and the remaining logarithmic, cubic, and Stirling terms have the displayed bounds. Within a cell, $|\log \varphi_\lambda(z) - \log \varphi_\lambda(z_j)| \leq CN^{-1/2}(1 + |z_j|)$. Consequently on $|z| \leq R_N = N^{1/10}$ the density ratio differs from one by at most $CN^{-1/5}$. Integrating this relative bound costs at most $CN^{-1/5}$. Outside this region,

$$\mathbb{E}\{(K - N\lambda + U)/\sqrt{N}\}^2 = \lambda + (12N)^{-1},$$

so Chebyshev's inequality bounds the step-density tail by CR_N^{-2} ; the same bound holds for the Gaussian tail. Thus

$$\|f_{N,\lambda} - \varphi_\lambda\|_{L^1} \leq CN^{-1/5}. \quad (10.3)$$

Translation to the centre $N\lambda^0$ changes neither the bound nor its uniformity. The finitely many small N are absorbed by increasing C .

For m independent coordinates, the total variation distance between products is at most the sum of the coordinate distances. This proves the same order for covariance $\text{diag } \lambda_N(h)$. To replace it by Λ , differentiate the normal density with respect to each variance on the fixed positive compact interval. The L^1 norms of these derivatives are uniformly bounded (they are the normal expectations of the absolute variance scores). Since $\|\lambda_N(h) - \lambda^0\| = O(N^{-1/2})$ uniformly, covariance replacement costs $O(N^{-1/2})$. This proves the asserted quantitative comparison.

Rounding $\sqrt{N}X_N^\circ + N\lambda^0$ to the nearest integer recovers each count almost surely, while independent jittering gives the reverse Markov kernel. On arbitrary Gaussian inputs the rounding kernel clips negative integers to zero, so it is defined on the entire observation space. In the limiting normal vector $X \sim N(B_*h, \Lambda)$, the statistic $Z = DX$ has law $N(h, \Sigma)$ and is sufficient: conditional simulation is

$$X = B_*Z + E, \quad E \sim N(0, \Lambda - \Lambda D^\top \Sigma^{-1} D \Lambda),$$

with E independent and its law independent of h . These kernels, the total variation approximation, and contraction of total variation under kernels prove the experiment comparison uniformly on the compact parameter set. In particular this proves the asserted local Le Cam comparison, not merely a central limit theorem. $\square$

Theorem 10.2 (Known-mark local information retained by contact scores). *With known mark laws and the preassigned centre and target submodel (10.1), the experiment retaining (10.2) is, uniformly on compact local parameter balls, asymptotically equivalent to*

$$T \sim N(F^\top \mathcal{L}h, F^\top \mathcal{L} \Sigma \mathcal{L}^\top F). \quad (10.4)$$

Its limiting information matrix for h is

$$I_{\text{contact}} = \mathcal{L}^\top (\mathcal{L} \Sigma \mathcal{L}^\top)^\dagger \mathcal{L} = \Sigma^{-1/2} P_{\text{ran}(\Sigma^{1/2} \mathcal{L}^\top)} \Sigma^{-1/2}. \quad (10.5)$$

It has rank r and kernel $\ker \mathcal{L}$. The contact-score reduction is locally asymptotically equivalent to the raw target subexperiment if and only if $\mathcal{L}$ is injective. Hence it retains all local moment information exactly when the within-contact polynomial products span V_{2n} .

Proof. Apply the fixed linear map $F^\top \mathcal{L} D$ to the total variation approximation in Lemma 10.1. Contraction gives convergence in total variation to (10.4), whose covariance is positive definite in its r coordinates. Identity kernels therefore compare these experiments in both directions with error tending to zero. The usual normal-shift likelihood gives the first expression in (10.5). For $M = \mathcal{L} \Sigma^{1/2}$, the identity $M^\top (M M^\top)^\dagger M = P_{\text{ran } M^\top}$ gives the second. Its rank and kernel follow immediately.

If $\mathcal{L}$ is injective, Z is recovered from $\mathcal{L}Z$ by a left inverse. Thus (10.4) is equivalent to $N(h, \Sigma)$, and Lemma 10.1 gives equivalence to the raw target subexperiment. Conversely choose distinct h_0, h_1 in a local parameter ball with $h_1 - h_0 \in \ker \mathcal{L}$. Their limiting contact distributions coincide, whereas $N(h_0, \Sigma)$ and $N(h_1, \Sigma)$ have strictly positive total variation distance. By the triangle inequality, any kernel purporting to reconstruct both raw limit laws has error at least half that distance. The same positive lower obstruction holds asymptotically by the uniform comparisons. The polynomial criterion is Theorem 2.2. $\square$

The uncentred contact block is $B_\nu(\theta) = \mathcal{L}_\nu \theta$. Its local fluctuation is precisely the linear type appearing in (10.2). Passing to inverse-Hessian contact scores applies the fixed invertible

block transformation $E_\nu \mapsto -B_{\nu,0}^{-1}E_\nu B_{\nu,0}^{-1}$, so it preserves the information statement. These are correlated raw-data score statistics, not independently noisy observations of distance values.

The centre λ^0 and the local submodel (10.1) are specified before observing the data. The theorem does not claim global sufficiency when both exposure and mark laws are unknown, or that a plug-in estimate of the centre automatically gives the same comparison. It supplies a genuine direct reduction and a complete local information-loss calculation in the stated full-dimensional target experiment. When $\mathcal{L}$ is injective, exact range-compatible score encoding does not introduce a γ^{-2} statistical variance penalty; independent measurement errors and numerical perturbations have the distinct stability behaviour of Theorem 7.2 and Appendix C.

10.1. Deterministic finite precision and the full exposure tangent model. The auxiliary randomization above prevents an exact real-valued encoding of the integer count vector. A finite-precision observation provides another, deterministic, way to remove that encoding.

Theorem 10.3 (Deterministic quantized score reduction). *Keep the known-mark target submodel and put $C = F^\top \mathcal{L}D$. Suppose $r \geq 1$. For a deterministic mesh $\eta_N > 0$, retain only*

$$Q_N = \left\lfloor \eta_N^{-1} C(N - N\lambda^0) / \sqrt{N} \right\rfloor \in \mathbb{Z}^r,$$

with the floor taken coordinatewise. If $\eta_N \rightarrow 0$ and $\sqrt{N}\eta_N \rightarrow \infty$, this experiment is asymptotically equivalent to (10.4). More precisely, for $0 < \eta_N \leq 1$,

$$\Delta(Q_N, \mathcal{T}) \leq C_0 \{N^{-1/5} + (\sqrt{N}\eta_N)^{-1} + \eta_N\}. \quad (10.6)$$

In particular $\eta_N = N^{-1/4}$ is admissible. The information and equivalence criterion of Theorem 10.2 therefore hold for this deterministic protocol as well.

Proof. Couple the unjittered score $S_N = C(N - N\lambda^0) / \sqrt{N}$ to $S_N^\circ = CX_N^\circ$ with the same counts. Their coordinatewise difference is at most $d_N = C_1 / \sqrt{N}$, deterministically. The two quantizers can differ only if a coordinate of S_N° lies within d_N of the grid $\eta_N \mathbb{Z}$. By Lemma 10.1, its joint law is within $CN^{-1/5}$ of the nonsingular normal law in (10.4).

For a univariate normal density f with variance in a fixed positive interval, the grid-boundary probability is at most Cd_N / η_N if $d_N \leq \eta_N / 2$. To see this uniformly in its mean, sum the supremum of f on each grid interval and use

$$\eta_N \sum_{j \in \mathbb{Z}} \sup_{[j\eta_N, (j+1)\eta_N]} f \leq 1 + \eta_N \int_{\mathbb{R}} |f'(x)| dx.$$

There are at most two boundary pieces per interval. A union bound over the r coordinates proves $\Pr\{\lfloor S_N / \eta_N \rfloor \neq \lfloor S_N^\circ / \eta_N \rfloor\} \leq C\{N^{-1/5} + d_N / \eta_N\}$. When $d_N > \eta_N / 2$ the asserted bound follows by increasing its constant. Quantization contracts total variation, so Q_N is within the same bound of a quantized Gaussian, uniformly in the local index.

The kernel adding an independent uniform point in the reported grid cell reconstructs the cell-average Gaussian density. Its L^1 distance from the Gaussian is at most $C_r \eta_N \int \|\nabla f(x)\| dx \leq C\eta_N$, by the integral mean-value inequality on cubes. The reverse kernel is deterministic quantization. Combining these kernels proves both deficiencies in (10.6). Randomization is used only by this comparison kernel, not by the observed statistic Q_N . The case $r = 0$ is the one-point experiment and needs no mesh. $\square$

An actual finite alphabet may be obtained by an overflow symbol outside a ball of radius $N^{1/10}$. Uniform second-moment bounds add only $O(N^{-1/5})$ to the comparison. Thus infinite digital precision is not hidden in the conclusion.

The least-favourable submodel has a precise relationship to the full exposure tangent experiment, rather than being an intrinsic physical distance observation.

Proposition 10.4 (Full exposure model and nuisance quotient). *For known marks and arbitrary bounded exposure perturbations $g \in \mathbb{R}^m$, set $\lambda_N(g) = \lambda^0 + g/\sqrt{N}$. The raw count experiment converges in Le Cam distance, uniformly on compact sets, to $X \sim N(g, \Lambda)$. In this Gaussian limit the transformation*

$$Z = DX, \quad E = (I_m - B_*D)X$$

*has independent components, with means $h = Dg$ and $\xi = (I_m - B_*D)g \in \ker D$. Quotienting the limit experiment by translations of this unrestricted nuisance gives exactly $Z \sim N(h, \Sigma)$; its contact compression has information (10.5). The least-favourable submodel is the section $\xi = 0$.*

Proof. The density proof of Lemma 10.1 applies unchanged to g in a compact set; jittering and rounding give both kernels. Since $DB_* = I_p$, $g = B_*h + \xi$ is a bijective linear parametrization by $h \in \mathbb{R}^p$ and $\xi \in \ker D$. The cross-covariance is

$$D\Lambda(I_m - B_*D)^\top = D\Lambda - \Sigma B_*^\top = 0.$$

The Gaussian components are therefore independent, the covariance of Z is Σ , and the law of $E - \xi$ does not depend on either parameter. Translation by $\ker D$ acts transitively on its nuisance means and leaves Z fixed. This proves the quotient assertion. It does not supply a kernel simulating an unknown ξ from Z , nor assert equivalence to the full experiment with that nuisance retained. The compact-set comparisons are made before this unrestricted Gaussian translation quotient; a compact nuisance set is not silently replaced by a translation group. $\square$

The centre remains an experimental input. A simple sufficient condition for replacing it by an independent pilot estimate is $\sqrt{N}\|\hat{\lambda}^0 - \lambda^0\| \rightarrow 0$ in probability, together with a fixed positive compact range for the pilot and continuity of the weights. On events where the scaled error is small, the centring shift is $o(1)$, the weights and Gaussian covariances converge, and the same uniform density bounds and mixture kernels apply. A root- M pilot gives this sufficient condition when $M/N \rightarrow \infty$. This is a sufficient, not an optimal, pilot regime. An ordinary pilot of size comparable to N does not meet it: its random centring error must then be included in the limit experiment. Nothing here asserts automatic plug-in sufficiency for unknown marks or an unrecorded pilot.

APPENDIX A. CLASSICAL MAXIMAL RANK AND MONOMIAL COMPARISON

The proof of the global codimension theorem does not invoke curve postulation. For comparison, the complete classical statement used for a single general space includes the injective range as well.

Proposition A.1 (Classical rational maximal rank). *For $4 \leq c \leq n+1$, a general c -plane $U \subset H^0(\mathbb{P}^1, \mathcal{O}(n))$ satisfies*

$$\text{rank}(\text{Sym}^2 U \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(2n))) = \min\left\{\binom{c+1}{2}, 2n+1\right\}.$$

The full-rank locus is a nonempty Zariski-open set defined over $\mathbb{R}$.

Precise classical reduction. Apply the maximal-rank theorem in [12, Theorem 1.2] to a general Brill–Noether curve with genus $g = 0$, degree n , ambient projective dimension $r = c - 1 \geq 3$, and restriction degree two. Its Brill–Noether number is $(r+1)(n-r) \geq 0$. A general c -dimensional subseries of $H^0(\mathbb{P}^1, \mathcal{O}(n))$ is base-point-free and embeds $\mathbb{P}^1$ in $\mathbb{P}^r$: its projection centre avoids the secant and tangent locus of the rational normal curve because $r \geq 3$. The space of such subseries is irreducible and parametrizes the general rational Brill–Noether map, up to the choice of a basis and source coordinates. Restriction of quadrics has source $\text{Sym}^2 U$ and target $H^0(\mathbb{P}^1, \mathcal{O}(2n))$, exactly the multiplication map above. This specializes the general theorem to the classical nonspecial rational range of [2, 3]. The complete series is immediate from monomials.

Nonvanishing maximal minors in real matrix charts gives the real assertion. This proof attributes, rather than reproves, the classical injective-range theorem. $\square$

For a monomial space $U_S = \text{span}\{t^a : a \in S\}$,

$$U_S^2 = \mathcal{P}_{2n} \iff S + S = \{0, \dots, 2n\}. \quad (\text{A.1})$$

This is the restricted additive 2-basis problem, distinct from an unrestricted polynomial subspace [17, 10, 11].

Proposition A.2 (Strict monomial obstruction and constructive bases). *If $n \geq 3$ and (A.1) holds, then $\binom{|S|+1}{2} \geq 2n + 2$. Thus no monomial normal space attains the native optimum at $(d, k) = (3, 8)$ or $(5, 11)$.*

For $a \geq 2$, $n \geq 2a$, write $n = ba + r$, $0 \leq r < a$. The set

$$S = [0, a - 1] \cup \{a, 2a, \dots, ba\} \cup [n - a + 1, n]$$

has $S + S = [0, 2n]$ and $|S| \leq 2a + \lfloor n/a \rfloor - 1$. In particular such bases exist with $|S| \leq 2\sqrt{2n} + O(1)$.

Proof. Representing $0, 1, 2n - 1, 2n$ forces $0, 1, n - 1, n \in S$. The distinct unordered pairs $(0, n)$ and $(1, n - 1)$ coincide in sum, so at least one of the $\binom{|S|+1}{2}$ products is redundant. At the displayed native optima the pair count is exactly $2n + 1$, respectively 15 and 21, making monomial recovery impossible.

For the construction, adding the multiples $0, a, \dots, ba$ to $[0, a - 1]$ covers $[0, ba + a - 1] \supset [0, n]$. Adding them to $[n - a + 1, n]$ covers $[n - a + 1, 2n - r]$. Adding the upper interval to itself covers $[2n - 2a + 2, 2n]$, which overlaps because $r \leq a - 1$. The last multiple belongs to the upper interval, giving the cardinality bound. Choose an integer nearest $\sqrt{n/2}$ for the asymptotic bound, handling the finitely many smaller values by the full interval. $\square$

For $k = 3d + 2$ one has the explicit family $S = [0, d] \cup [2d + 1, 3d + 1]$, with complementary exponents $E = [d + 1, 2d]$. The three sum intervals are $[0, 2d]$, $[2d + 1, 4d + 1]$ and $[4d + 2, 6d + 2]$. Each column of the interpolation matrix J has one strict coordinatewise sign, as follows by expanding $d_i \prod_{a \neq i} (t - r_a)$ with positive roots. In particular a signed Je_{d+1} is positive. Taking $T = J \text{span}\{e_a : a \in E\}$ realizes the monomial complement locally by Theorem 6.1; generic small loading perturbations give globally separated full-product realizations. The unrestricted fibre dimension for this family is $5d(d + 1)/2$. The smaller sets $\{0, 1, 3, 5, 6\}$ and $\{0, 1, 2, 5, 8, 11, 12, 13\}$ also satisfy (A.1); no optimal restricted-basis constant is claimed.

For a nonorthonormal normal frame $W = ZO$, set $K = W^\top W$ and $G = \frac{1}{2} W^\top D^2 C_R W$. The exact dual readout is

$$KG^{-1}K = W^\top RW. \quad (\text{A.2})$$

Indeed $K = O^\top O$ and $G = O^\top B^{-1}O$, and direct inversion proves the identity. If W has columns $J^{-\top} e_a$ for $a \in S$, its readout entries are θ_{a+b} . Dropping the Gram factors in (A.2) would give an incorrect normalization.

APPENDIX B. FINITE CALIBRATED SAMPLES AND OBSERVED EXPOSURES

Keep the roots, clocks and calibrated tables fixed and separated. Put $\delta = M_1 - M_2 \neq 0$, assume $\pi_j \in [\varepsilon, 1 - \varepsilon]$ with $0 < \varepsilon < 1/2$, and define

$$\kappa(u) = u^2(1 - u)^2 \sum_{a=1}^4 \frac{\delta_a^2}{M_{2a} + u\delta_a}, \quad g(u) = 1/\kappa(u).$$

On this interval g is Lipschitz with a finite constant L_g . We distinguish a chosen target-design weight λ from the realized sample allocation n_j/N . The theorem below estimates the metric

indexed by the chosen known λ . It estimates the realized normalized sample information only when those weights are chosen as n_j/N , and the limiting information when $n_j/N \rightarrow \lambda_j$.

Theorem B.1 (Fixed-design categorical recovery). *Observe $n_j \geq 1$ independent marks at clock j with law p_j , independently across groups. For empirical cell frequencies $\bar{p}_j$, set*

$$\hat{\pi}_j = \text{proj}_{[\varepsilon, 1-\varepsilon]} \frac{\langle \bar{p}_j - M_2, \delta \rangle}{\|\delta\|_2^2}, \quad W_{\cdot j} = \frac{\rho_j}{\lambda_j} w_{2n}(T_j), \quad \hat{\theta} = W(g(\hat{\pi}_j))_j.$$

Then $\hat{\theta} \in \Omega$ and

$$\mathbb{E}\|\hat{\theta} - \theta\|_2^2 \leq \frac{\|W\|_{\text{op}}^2 L_g^2}{\|\delta\|_2^2} \sum_j n_j^{-1}. \quad (\text{B.1})$$

On compact interior design sets, the induced estimates of R and the inverse normal blocks have squared risk $O(\sum_j n_j^{-1})$. If $\mathcal{L}$ is injective, inversion of those induced jets returns $\hat{\theta}$ exactly.

Proof. Before clipping, the estimate of π_j is unbiased. For a one-hot cell vector X , $\text{Var}\langle X, \delta \rangle \leq \|\delta\|_2^2$. Projection cannot increase distance to the true point in the interval, so $\mathbb{E}|\hat{\pi}_j - \pi_j|^2 \leq (n_j \|\delta\|_2^2)^{-1}$. Apply the Lipschitz bound for g and the operator norm of W . All estimated moment weights are positive. On compact design sets they lie in a fixed positive box; the moment image is a compact convex subset of Ω . The map to R and positive block inversion are Lipschitz there. Finally the induced inverse jets satisfy $\hat{G}_\nu^{-1} = \mathcal{L}_\nu \hat{\theta}$, proving the last assertion. $\square$

The bound allows the π_j to be coupled by the polynomial model and does not assert a full-dimensional fixed-design likelihood. Its constants need not stay bounded when roots or clocks collide, or probabilities approach the boundary.

Theorem B.2 (An additional observed-exposure protocol). *Observe independent counts $N_j \sim \text{Pois}(N\lambda_j)$ and, conditionally, N_j categorical marks of law p_j . Suppose $0 < \lambda_- \leq \lambda_j \leq \lambda_+$. Clip N_j/N to this interval, use the preceding clipped mark estimate, and use any value in $[\varepsilon, 1-\varepsilon]$ when $N_j = 0$. Substitution into (3.2) gives uniform squared risk at most K/N on fixed compact designs.*

An open exposure box has an open image in the moment cone. If the mark laws are known and exposures are interior, the plug-in moment estimate has limiting covariance

$$\Sigma_\theta = \sum_j \frac{\rho_j^2}{\lambda_j^3 \kappa_j^2} w_{2n}(T_j) w_{2n}(T_j)^\top > 0 \quad (\text{B.2})$$

after multiplication of its error by $\sqrt{N}$. This is the efficient functional covariance in that known-mark-law subexperiment; its squared-risk order N^{-1} is locally unimprovable.

Proof. Clipping gives $\mathbb{E}|\hat{\lambda}_j - \lambda_j|^2 \leq \lambda_+/N$. For $X \sim \text{Pois}(a)$,

$$\mathbb{E}[X^{-1}; X > 0] \leq 2\mathbb{E}[(X+1)^{-1}] = 2(1 - e^{-a})/a, \quad \Pr(X = 0) = e^{-a} \leq 1/a.$$

Conditioning on the counts therefore gives mark-estimation risk $O(N^{-1})$. The weight functions $\rho_j/(\lambda \kappa(u))$ are uniformly Lipschitz on the fixed positive compact rectangle, proving the risk bound.

The exposure-to-moment Jacobian has columns $-\rho_j w_{2n}(T_j)/(\lambda_j^2 \kappa_j)$ and full row rank. This gives the open-image assertion. The Poisson central limit theorem and the delta method give (B.2). The exposure information per unit N is $\text{diag}(1/\lambda_j)$; its inverse pushed forward by this Jacobian is exactly the displayed covariance. The regular local information bound for smooth functionals, or the normal target experiment in Lemma 10.1, gives efficiency and the lower risk order [16]. $\square$

Counts are genuinely observed in this additional protocol. Unknown exposures cannot be inferred from fixed-count marks alone. Jointly unknown marks and exposures need not have the known-mark-law efficient covariance (B.2).

APPENDIX C. COMPUTED FINITE-OFFSET FEATURES AND INDEPENDENT NOISE

At a globally separated informative point b_0 , choose the normal frame Z and the q vectors v_a consisting of its columns and their pairwise sums. Quadratic evaluation is an isomorphism $\mathcal{Q} : \text{Sym}_c \rightarrow \mathbb{R}^q$. Define

$$\mu_h(\theta)_a = \frac{1}{2} \{C_{R_\theta}(b_0 + hv_a) + C_{R_\theta}(b_0 - hv_a)\}.$$

Fix a compact convex $\Theta \subset \Omega$ containing the true parameter and the clipped plug-in estimates. For sufficiently small $h > 0$ all targets are positive and lie in the common analytic projection tube. Symmetric Taylor expansion, also differentiated in θ , gives

$$h^{-2}\mu_h(\vartheta) = \mathcal{Q}((\mathcal{L}\vartheta)^{-1}) + h^2 r_h(\vartheta), \quad \sup_{0 < h \leq h_0} \|r_h\|_{C^1(\Theta)} < \infty. \quad (\text{C.1})$$

Theorem C.1 (Computed finite-offset encoding and inversion). *For a raw estimator with squared risk at most K/N , compute $\hat{\mu}_h = \mu_h(\hat{\theta})$ and set*

$$\tilde{\theta}_h = \mathcal{L}^\dagger [\mathcal{Q}^{-1}(h^{-2}\hat{\mu}_h)]^{-1}.$$

For small enough h all inverses exist and

$$\mathbb{E}\|\tilde{\theta}_h - \theta\|^2 \leq 2K/N + 2C^2 h^4 / \gamma^2.$$

For a fixed design, $h_N = o(N^{-1/4})$ makes this output equivalent to the raw estimator at scale $\sqrt{N}$. With uniform tube and metric bounds along varying designs, it suffices that $\sqrt{N}h_N^2/\gamma_N \rightarrow 0$. A deterministic contact-computation error at most τ_N adds parameter error at most $C'\tau_N/(\gamma_N h_N^2)$, provided positive inversion remains valid.

Proof. Smooth positive inversion in (C.1) gives

$$F_h(\vartheta) := [\mathcal{Q}^{-1}(h^{-2}\mu_h(\vartheta))]^{-1} = \mathcal{L}\vartheta + h^2 E_h(\vartheta), \quad \|E_h\|_{C^1} \leq C.$$

Thus $\tilde{\theta}_h = \hat{\theta} + h^2 \mathcal{L}^\dagger E_h(\hat{\theta})$ and its extra error is at most Ch^2/γ . Squaring proves the risk and scaling claims. An error τ_N perturbs the argument of inversion by at most $\|\mathcal{Q}^{-1}\|\tau_N/h_N^2$; use the inverse identity in Theorem 7.2. $\square$

This theorem concerns deterministic re-encoding of an already available estimator. It is not an independent observational route or a new minimax rate. Its offset condition controls approximation bias; the leading statistical risk is inherited from the raw estimate. The direct score reduction of Section 10 does not use this construction.

For comparison, consider the different experiment of independent noisy readings

$$Y_i = \mu_h(\theta) + \sigma \xi_i, \quad \xi_i \sim N(0, I_q), \quad i = 1, \dots, N.$$

For compact convex Θ and sufficiently small fixed h , the exact finite-offset mean satisfies

$$ah^2\|\theta - \theta'\| \leq \|\mu_h(\theta) - \mu_h(\theta')\| \leq bh^2\|\theta - \theta'\|$$

for $a, b > 0$. Indeed $F_h = \mathcal{L}\theta + h^2 E_h$ and $h^2\|DE_h\| \leq \gamma/2$ give a lower bound by the triangle inequality and the mean-value bound for E_h ; positive inversion and $\mathcal{Q}$ transfer it to μ_h . The upper bound follows directly from (C.1). Fitting the exact mean by least squares consequently gives

$$\sup_{\theta \in \Theta} \mathbb{E}\|\hat{\theta}_N - \theta\|^2 \leq \frac{4q\sigma^2}{Na^2h^4},$$

since the fitted mean is within twice the sample-mean error of the true mean. At an interior parameter the information is $I_h = \sigma^{-2} D\mu_h^\top D\mu_h > 0$, the asymptotic covariance is I_h^{-1} , and the local asymptotic minimax risk is $\text{tr}(I_h^{-1})/N$ [16]. The least eigenvalue of I_h/h^4 is comparable to γ^2/σ^2 under the fixed metric bounds, by differentiating (C.1). Gaussian tails and the lower Lipschitz bound justify convergence of local risks. A quadratic-truncated fit at fixed h would instead generally retain an $O(h^2)$ bias and is not used. The h^{-4} variance in this independent-noise experiment must not be transferred to the correlated computed or raw-score statistics.

APPENDIX D. COMPLETE CONTACT, DESIGN FIBRES, AND ENDPOINT REPRESENTATIONS

These statements concern different observation categories from a prescribed normal block. We record their mathematical interfaces and proofs so that the one-contact and multi-contact results do not erase the complementary conclusions.

D.1. Complete conormal ambiguity. Suppose $\mathcal{A}_A : \text{Sym}_d \rightarrow \mathbb{R}^k$, $S \mapsto (a_i^\top S a_i)_i$, is injective. The complete conormal union of the rank-one image is dense in the inverse image, under $\mathcal{A}_A^*$, of the singular symmetric matrices: a conormal n at $q_A(\eta)$ satisfies $(\sum_i n_i a_i a_i^\top) \eta = 0$. Conversely a real singular matrix has a nonzero real kernel vector. For $d \geq 3$, no nonzero quadratic form vanishes on all singular symmetric matrices. To see this, restrict a quadratic polynomial to matrices diagonal in any fixed orthogonal basis. Every monomial of degree at most two omits at least one of the d diagonal variables, so vanishing on every coordinate hyperplane forces the restriction to vanish identically. Every real symmetric matrix is orthogonally diagonalizable, so the polynomial is zero. For $d = 2$, direct substitution of rank-one matrices leaves exactly the determinant line. This is the elementary coordinate consequence of classical Veronese duality [8].

Under the surjective map $\mathcal{A}_A^*$, pure kernel and mixed quadratic terms must also vanish, since singular symmetric matrices span Sym_d . Thus the complete-conormal ambiguity is zero for $d \geq 3$ and one-dimensional for $d = 2$. In the latter case it is generated by $B_{ii} = 0$, $B_{ij} = \det(a_i, a_j)^2$ and is intersected with the admissible native positive cone. In root coordinates the native linear span consists exactly of symmetric matrices satisfying

$$\frac{(r_j - r_i)X_{ij}}{d_i d_j} - \frac{(r_l - r_i)X_{il}}{d_i d_l} + \frac{(r_l - r_j)X_{jl}}{d_j d_l} = 0 \quad (i < j < l). \quad (\text{D.1})$$

The divided-difference identity for $(t - r_i)^{-1}(t - r_j)^{-1}$ proves these relations for $JH(\theta)J^\top$. Conversely they leave k free diagonal entries and $k - 1$ off-diagonal difference coordinates, hence dimension $2k - 1$, equal to the Hankel span. Testing B in (D.1) gives the exact two-parameter transversality criterion. Equality of second jets here is not equality of all higher contact terms.

D.2. Exact dual-jet query count. In a deterministic exact-real oracle whose scalar answers are values of the dual normal quadratic forms, full product rank requires exactly $p = 2k - 1$ scalar queries. Squares span within-contact products by polarization, so select p independent square functionals for sufficiency. For necessity, follow any adaptive transcript at an interior parameter. Fewer than p scalar linear answers have a nonzero common kernel. A sufficiently small perturbation in that kernel remains admissible and reproduces the same answers, hence the same adaptive choices, by induction. This is a query statement about the specified dual oracle, not a statistical sample bound or a count of ordinary scalar distance readings.

D.3. Reciprocal design fibres. Put $a_j = \rho_j/\kappa_j > 0$, let V_m be the $p \times m$ moment matrix, and use reciprocal weights $\lambda_j = a_j/\omega_j$. Define on the open moment cone

$$\chi(\theta) = \min_{V_m \omega = \theta, \omega > 0} \sum_j a_j / \omega_j.$$

The zeroth moment bounds every fibre. The objective tends to infinity at its boundary and is strictly convex, so there is a unique minimizer. Its diagonal Hessian $D_\omega > 0$ and differentiation of the constrained first-order equations give

$$D^2\chi = (V_m D_\omega^{-1} V_m^\top)^{-1} > 0.$$

Indeed $D_\omega d\omega = V_m^\top d\nu$ and $V_m d\omega = d\theta$, while ν is the gradient of the value. The implicit function theorem also gives smoothness. Here $m = p + 2$, so the affine fibre has dimension two. Every ray from its minimizer meets a higher level once, because the restriction is strictly convex and diverges at the positive-fibre boundary. Its higher levels are smooth circles. Consequently the unit-exposure-budget image is $\{\theta : \chi(\theta) \leq 1\}$, with singleton boundary fibres and circular interior fibres. Homogeneity $\chi(t\theta) = t^{-1}\chi(\theta)$ makes the boundary regular relative to the open cone. Injective contact coordinates transport this geometry without changing the fibres.

D.4. Endpoint value-graph representations. For a labelled semialgebraic value graph specified over algebraic coefficients, fix a hidden dimension and a specified normalization of a positive block matrix. Every representation

$$F(x) = \min_{z \geq 0} (x, z)^\top \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} (x, z)$$

is characterized exactly by positivity, that normalization, and the quantified relations

$$z \geq 0, \quad Cz + B^\top x \geq 0, \quad z_i(Cz + B^\top x)_i = 0,$$

together with the value equality for every labelled x . Strict convexity in z makes these Karush–Kuhn–Tucker conditions necessary and sufficient. Real quantifier elimination therefore computes the representation fibre at the fixed hidden dimension [4]. Enumeration over dimensions terminates under a finite-representability promise, yielding the first nonempty normalized fibre; no unconditional finite hidden-dimension bound is implied. This input is a scalar value graph, not an optimizer map as in inverse parametric programming [9, 14].

D.5. Deterministic local root costs. The native expansion $p(u, v) = p_0 + L_p u + D_p v + O((\|u\| + \|v\|)^2)$ gives the information in Proposition 3.1. With squared Hellinger distance $h^2(p, q) = \sum_e (\sqrt{p_e} - \sqrt{q_e})^2$, profiling the free coordinates yields the quadratic efficient cost after multiplying by $4/t^2$. Comparing a physical target variance tb , $b \geq 0$, with synchronized $v = tq_A(\eta)$ gives (1.1). Full rank of A gives coercivity and bounded scaled minimizing sequences, making the Taylor remainder uniform on the relevant local chart. Physical splitting has size $\sqrt{t}$. For C^2 nonlinear loadings g_i with $g_i(0) = 0$ and $Dg_i(0) = a_i^\top$,

$$g_i(\sqrt{t}\eta)^2 = t(a_i^\top \eta)^2 + O(t^{3/2})$$

uniformly on bounded η . These local statements do not require injectivity of $\mathcal{A}_A$ on all of Sym_d ; global chart selection uses the separated-fibre hypotheses. At $t = N^{-1/2}$ the same cost is the physical local Gaussian profile of Theorem 9.1.

REFERENCES

- [1] R. Balan, P. Casazza, and D. Edidin, On signal reconstruction without phase, *Appl. Comput. Harmon. Anal.* **20** (2006), 345–356.
- [2] E. Ballico and Ph. Ellia, The maximal rank conjecture for nonspecial curves in $\mathbb{P}^3$, *Invent. Math.* **79** (1985), 541–555.
- [3] E. Ballico and Ph. Ellia, The maximal rank conjecture for non-special curves in $\mathbb{P}^n$, *Math. Z.* **196** (1987), 355–367.
- [4] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second ed., Springer, 2006.
- [5] A. Conca, M. Mostafazadehfard, A. K. Singh, and M. Varbaro, Hankel determinantal rings have rational singularities, *Adv. Math.* **335** (2018), 111–129; arXiv:1709.05685, Sections 1–2.

- [6] D. Eisenbud, C. Huneke, and B. Ulrich, Heights of ideals of minors, *Amer. J. Math.* **126** (2004), 417–438; arXiv:math/0209184.
- [7] C. J. Geyer, On the asymptotics of constrained M -estimation, *Ann. Statist.* **22** (1994), 1993–2010.
- [8] I. M. Gelfand, M. M. Kapranov, and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, 1994.
- [9] A. B. Hempel, P. J. Goulart, and J. Lygeros, Inverse parametric quadratic programming and an application to hybrid control, *IFAC Proceedings Volumes* **45**(17) (2012), 68–73.
- [10] J. Kohonen, A meet-in-the-middle algorithm for finding extremal restricted additive 2-bases, *J. Integer Seq.* **17** (2014), Article 14.6.8.
- [11] J. Kohonen, An improved lower bound for finite additive 2-bases, *J. Number Theory* **174** (2017), 518–524.
- [12] E. Larson, The maximal rank conjecture, arXiv:1711.04906, Theorem 1.2.
- [13] B. Mourrain and V. Y. Pan, Multivariate polynomials, duality, and structured matrices, *J. Complexity* **16** (2000), 110–180.
- [14] N. A. Nguyen, S. Olaru, P. Rodriguez-Ayerbe, M. Hovd, and I. Necoara, Constructive solution of inverse parametric linear/quadratic programming problems, *J. Optim. Theory Appl.* **172** (2017), 623–648.
- [15] C. Scheiderer, Extreme points of Gram spectrahedra of binary forms, *Discrete Comput. Geom.* **67** (2022), 1174–1190.
- [16] A. W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998.
- [17] G. Yu, Upper bounds for finite additive 2-bases, *Proc. Amer. Math. Soc.* **137** (2009), 11–18.
- [18] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [19] M. Green and R. Lazarsfeld, On the projective normality of complete linear series on an algebraic curve, *Invent. Math.* **83** (1986), 73–90.
- [20] J. A. Eagon and D. G. Northcott, Ideals defined by matrices and a certain complex associated with them, *Proc. Roy. Soc. London Ser. A* **269** (1962), 188–204.
- [21] A. R. Kustin, Canonical complexes associated to a matrix, arXiv:1509.04367, 2015, discussion following (1.0.2) and Section 2.9.
- [22] G. Kempf and D. Laksov, The determinantal formula of Schubert calculus, *Acta Math.* **132** (1974), 153–162.
- [23] W. Fulton, *Intersection Theory*, second ed., Springer, 1998, Theorem 14.4.